

Chapter 5

THE MANIPULATION OF LARGE TUBING

General Observations

We apply the term large tubing to tubing with an external diameter of about 35 mm and greater. The problems encountered in working this size of tubing differ from those connected with the working of smaller tubes. These problems usually arise from either the greater weight of the tubing or from the greater heat losses from the site of working by conduction along the glass and by radiation from its surface. The weight poses no problem that practice in manipulation and the use of supports or rollers cannot overcome, but the increased heat losses cause the tube to become too hot to hold for a considerable distance along its length, and the radiation from the large mass of hot glass may prevent the tubing from being manipulated with unprotected hands.

The obvious solution to the above problems is to protect the hands with asbestos gloves, but even this has its disadvantages. Asbestos gloves prevent the same degree of manipulative skill as can be attained without them, which increases the difficulty of working the heavy tubing, and if the radiation is great the air inside the glove may become unbearably hot. For the amateur glass-blower it is frequently more convenient to clamp the tubing and carry out the work with hand torches.

Many operations with large tubing can be performed easily with a glass-working machine (Chapter 6). The following pages describe some methods for hand-working which are applicable to hard borosilicate glass tubes, e.g. *Pyrex* and *Phoenix*.

Bending Big Tubing

Method for the More Skilled Worker—Shrinking down a Short Portion

Bends cannot be made in tubing of greater diameter than 12 mm without blowing. Skilled workers can bend tubing up to about 25–30 mm diameter by heating it with rotation over a length equal to about twice the diameter, shrinking it over this length to about half

BENDING BIG TUBING

its diameter until the glass is beginning to sag, removing it from the flame, bending and blowing it out with one puff. The unskilled glass-blower has, however, to rely on heating and bending a short portion at a time.

Making Sharp Bends and Bending a Portion at a Time

A good method of producing the desired result is that of sealing two separate pieces of glass held at the required angle, or, in effect, making a T-seal at the end of a piece of tubing. This, in fact, is the only way, for the unskilled worker, of producing good sharp bends in tubing greater than 15–20 mm diameter.

The method of heating and bending a portion at a time will give a satisfactory bend whose radius, for the outside wall of the bend, is between five and six times that of the tube. The method has been used satisfactorily on tubes up to 50 mm diameter. The tube, closed at one end and with a blowing tube attached at the other, is clamped horizontally near one end, and heated, where the bend is required to start, over a length equal to about twice the tube diameter, with a large but not too hot flame from the hand torch. As the glass heats up it will begin to bend under its own weight, and the free end is supported until the glass reaches a dull red heat when it is allowed to fall a short way but prevented from sagging too far. The glass on the outside of the bend tends to flatten and it is heated fairly strongly with a hotter flame and blown out evenly. Kinks on the inside of the bend are heated even more strongly, first with a pin-point flame until they shrink in, and then with a larger flame which enables them to be blown out smoothly. The clamp is then rotated in its boss until the free end of the tube is again horizontal, and the procedure is repeated on the neighbouring portion and so on until the required bend is obtained.

With tubing up to about 30 mm in diameter the whole bend can be annealed by flame at the finish, but above this diameter it is advisable to anneal each portion for a few minutes after it is bent and shaped, and before one moves on to the next portion. With extra large tubing it is better to have two workers, one on either side of the tube, each with a hand torch.

Drawing Off Large Tubes

The need for a strong spindle is avoided if the end of the tube is to be blown out to make a joint or if it is to be drawn off by way of cutting the tube. In this case the tube may be drawn off under less well controlled conditions than if a strong spindle is required. It is possible to clamp the tube, preferably in a vertical position, and work

THE MANIPULATION OF LARGE TUBING

the glass with hand torches. It may be necessary for two persons to carry out this operation so that the lower part of the tube can be supported to prevent too rapid drawing off as it falls under gravity. The lower part of the tube will remain much cooler than if the glass were worked in a horizontal position.

Drawing large diameter tubing off to spindles can be accomplished by a simple extension of the methods previously described. It is almost essential to support the tube on a stand or set of rollers since the great weight of a relatively long length of tubing will prevent accurate rotation of the two pieces. It may also be found that the normal bench torch does not produce a hot enough flame for the purpose. If this is the case it is possible to arrange two hand torches in laboratory clamps so that the tubing is rotated between them. Then, if the tubing is not drawn down too rapidly, a strong spindle will be produced. Asbestos gloves may be necessary to protect the hands from the hot glass and the radiation if the tube is to be kept in the flame for long enough to collapse down sufficiently to form a strong spindle.

Cutting a length of tubing can be a useful initial stage in putting in a strong spindle. This reduces an unwieldy length of heavy tubing to a more manageable size which can be held in one hand while a handle is put in preparatory to drawing off the end to a strong spindle. If an uneven end has been produced by faulty cutting and it is not intended to draw the tube off, the end may be tidied up considerably by local working with a hand torch and the use of forceps and scissors for removing excess glass. Any glass so worked should be annealed carefully to prevent cracking on cooling.

Closing Large Flasks

The necks of large flasks (about 5 litres capacity) are usually of thick glass, and they must be heated slowly. Before the top is closed a side arm must be put on the flask, and the neck is usually the best place for it. The mouth of the flask is closed with a rubber stopper wrapped in asbestos paper and carrying a blowing tube. A considerable area of the neck of the flask is slowly heated, using initially a large smoky flame and finally a hot flame. When the flame is tinged yellow a hole is blown in the side of the neck, of the right size to fit the side arm. The end of the side arm tube should previously have been thickened by heating, flanging and running down, otherwise a bad joint will be made because the thin, side arm glass runs so much more easily than the thick glass of the flask neck. The side arm is joined to the neck by the normal T-piece procedure, and the neck all round the joint is annealed carefully. The top of the neck is then heated and pushed in

CLOSING LARGE FLASKS

with a spike until a handle can be attached as in Figure 32, *I*. The neck is pulled off near the top, giving a spindle which will probably be very thin (Figure 32, *II*). This is removed and the end rounded by running down and blowing out a number of times—to blow out all at once requires a very large flame, and the end must be blown out piece by piece with all ordinary flames.

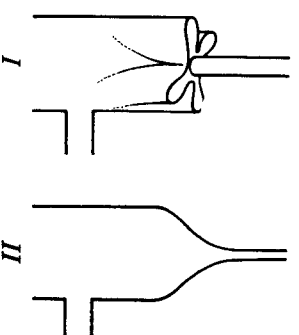


Figure 32. Closing the neck of a large flask

Joining Big Tubes

Once again the methods previously described for joining two tubes are applicable provided the limitations imposed by the size of the tubing are borne in mind. Joining two large diameter (say 4 cm) tubes in the flame of a bench torch follows the same course as joining smaller tubes by the same method. Care must be taken to see that the tubes are long enough not to become too hot to hold. Tubing of 4 cm diameter may become too hot to hold for about ten inches on either side of the site of working, since not only is more heat conducted along the glass but also a much larger flame is necessary.

Working large diameter tubing clamped is very often more convenient than attempting to work it with a bench torch. Joining two large diameter tubes by this method follows the same pattern as the joining of smaller tubes in similar circumstances: first an adequate supply of tools and thin glass rod to meet any emergency is collected. Then the two tubes should be prepared by blowing out the ends. It is best to blow holes of only about three quarters of the diameter of the tube (Figure 33) as the ends of the drawn-off tubes are liable to have rather thin walls and this allows them to be thickened as the joint is made. The tubes are heated and pushed together as before, or, if this is not possible, the flanges of glass left by blowing out the ends are heated and pushed together with a spike, or the gap is filled with glass

THE MANIPULATION OF LARGE TUBING

rod. While this is being done it is necessary to heat the joint more or less evenly and not just on one side. If this is difficult two operators, each with a hand torch, may be found an advantage.

All other operations—the working of the glass by alternate collapsing and blowing, sealing pinholes, and finally blowing the joint to an acceptable shape—are the same as previously described. The flow of glass under gravity will be more noticeable when large tubes are joined, principally because the time required to make the joint is greater. This flow can be combated as before.

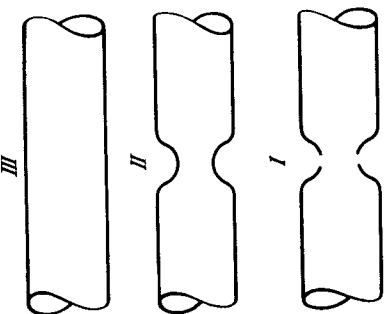


FIGURE 33. Joining large tubing

The final annealing of the joint and adjacent tubing is of the utmost importance with large tubes. If this is neglected there is a high probability that the joint will crack as a result of the strains induced by working and cooling the glass.

Big T-Joints

Large sized T-joints can be made using a bench torch and manipulating the glass in precisely the same way as previously described. This is difficult since the whole weight of the glass must be carried by the hands, because the way in which the tubing must be turned in the flame precludes the use of supports. The simplest way of making these joints is to clamp both tubes in such a way that they can be pushed together at the required time.

The two parts should be prepared by blowing a hole in the end of the side arm smaller than the bore of the tube and then blowing a similar sized hole in the cross-piece. These parts should next be clamped so that the tubes are nearly touching and can readily be

BIG T-JOINTS

pushed together. The two tubes are then heated as uniformly as possible around the join, and when the glass has become soft the two tubes are pushed together. If they do not make contact at any point it may be possible to push the glass in this region together with a spike or else to bridge the gap with thin rod. The joint is then worked, by the usual method, to fuse the two tubes together. When the joint is blown out to its final form it is advisable to heat a fairly large segment of it at a time; and blowing the glass carefully while it is in the flame very often leads to a better finish. Mending any pinholes in the joint is much more easily accomplished at this stage and therefore a careful inspection should be made before the final annealing, which should be carried out thoroughly. For work on very large diameter tubing it may be necessary to use two hand torches for the above operations.

Very often it is found more convenient to replace a right-angle bend by a T-joint made right at the end of the drawn-off cross-piece. This makes a very neat right-angle join in tubing which has too large a diameter to bend easily.

Big Internal Seals

Big internal seals can be made by any of the methods described in the previous chapter although by far the easiest method is to work with the glass clamped and to keep the internal tube in one piece.

Preparation of the materials is one of the most important phases of the operation, since once the glass-blowing is started it is virtually impossible to leave the work until it is finished and annealed thoroughly. The inner tube should be prepared as before with a slight ridge round the tube at the position of the joint. The outer tube should then be closed off to a round end and the hole to take the internal tube blown. This should be left with a slight rim of glass and not flame-polished. Before this can cool the internal tube should be inserted and clamped in position and work started on the joint. If the glass is not in good contact initially it may be heated and pushed on to the internal tube with a spike. Care should be taken in so doing to see that the flame is not concentrated too much on one side of the partly made joint, allowing the other side to cool and crack—one of the dangers of pushing the glass in with a tool. The attention is concentrated on the localized area being worked and the rest of the join is allowed to cool, thus straining the glass excessively. The localized working itself tends to strain the joint, but if care is taken to see that the whole joint is kept hot the strain is dissipated by a continuous annealing process.

Once the two tubes have been pushed into close contact and the

THE MANIPULATION OF LARGE TUBING

joint is air-tight it can be finished as previously described. These internal seals require very thorough annealing and it is best to heat them to a dull red heat for a short while before cooling slowly in the flame. The use of two torches facilitates uniform heating during the annealing process, and since no other operations are necessary one can be held in each hand if they are both prepared before the commencement of the glass-blowing. This use of two flames is probably unnecessary except for very large tubing.

Mending Cracks in Large Tubing

When large tubing has been worked and has been annealed in a flame it is usually strained to a considerable extent, which sometimes causes it to crack. Mending these cracks without spreading them is an operation requiring a considerable amount of patience. It is one of the cases in which it is necessary 'to make haste slowly'. The use of two flames is an advantage as they allow the heating up of both ends of the crack at the same time. The glass should initially be warmed uniformly and slowly with small cool flames. When it has started to heat up, the rate of heating is increased, concentrating the flame slightly on the ends of the crack so that these parts warm up just a little faster than the rest of the glass. Once the glass has heated up sufficiently to flow slowly the crack is allowed to seal itself, using only a cool flame as previously described. Finally, when a complete repair has been effected, the glass is annealed.

Mending Cracks near Complex and Large Seals

The above remarks apply equally here, except that extreme care is needed to prevent the crack spreading into the complex region. If the crack spreads to an internal seal and the inner tube cracks, it will be virtually irreparable. This spreading can be prevented if the whole of the complex joint is heated at the same time and at the same rate as the glass around the crack. The heating must be slow to allow time for the inner parts of the joint to be heated adequately by conduction.

Chapter 6

SOME OPERATIONS WITH A GLASS- WORKING MACHINE

We have mentioned (p. 2) the difficulty of holding and rotating a piece of glass tube in each hand and operating on a semi-molten mass between the two pieces. A very satisfactory way of resolving this problem is to use a machine—the *Edwards Model G 3 Glass-Working Machine* (made by Edwards High Vacuum Ltd) is of great value. The glass to be worked is held in two chucks which grip the tubing and rotate it synchronously at a speed which is variable over a considerable range (120–500 rev/min). One of the chucks can be moved towards or away from the other. The glass may also be held in only one chuck.

The glass can be worked both by blowing into it as it rotates and by the use of tools which can be mounted in various ways or held in the hand. Thus operations involving the production of articles with circular symmetry can be carried out with great ease, and a skilled operator is by no means necessary. The advantages of using this machine, when possible, for work involving tubing exceeding 3–4 cm in diameter are, we think, particularly manifest. The operation of putting a spindle in a large tube, for example, is done by hand only with difficulty, whereas with the machine it can be done rapidly and effectively. In this Chapter we attempt to give merely a general idea of how such a machine can be used; the detailed manipulations involved soon become clear after a little practice. The machine is valuable either for the production of special parts in research and development work or for continuous operation on repetition work by unskilled personnel.

General Description

Two views of the glass-working machine are shown in FIGURES 34 and 35. It consists of a substantial cast base fitted with a fixed headstock and a movable tailstock. This can be moved towards or away from the headstock by operating the tailstock traverse control. Rotating mandrels in the headstock and tailstock are synchronously driven; the tailstock mandrel drive can be disconnected and locked

SOME OPERATIONS WITH A GLASS-WORKING MACHINE

in position so that the mandrel does not rotate, and it can then be employed as a tool holder for tools used to work single lengths of tubing held in the headstock chuck. The tool can be moved towards or away from the glass using the tailstock traverse as usual.

In the front of the machine there is a burner mounting bar which normally carries either a cannon burner, operating with gas and air or gas-air-oxygen mixtures, or a crossfire burner, operating with gas-oxygen mixtures. Both burners can be used together and their flames directed on the same region. The burner mounting bar has a

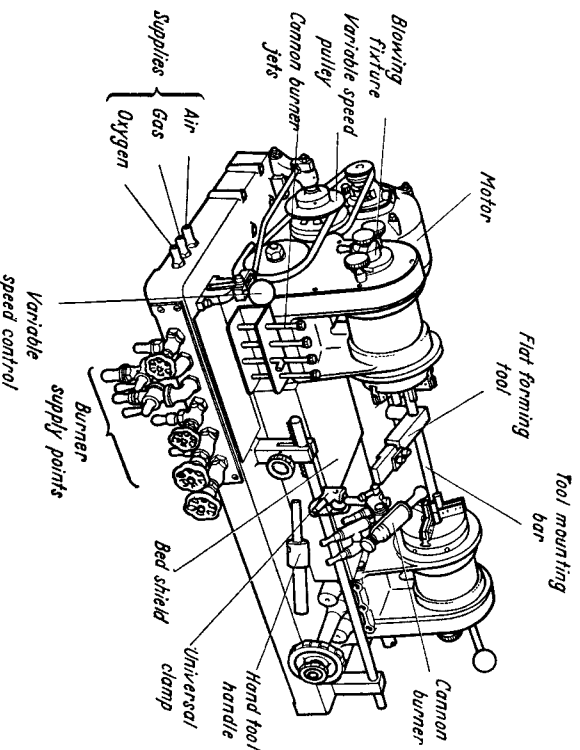


FIGURE 34. A glass-working machine

rack-and-pinion mechanism so that the burner or burners can be traversed a limited distance along the glass and be made to follow up the work.

At the rear of the machine there is a tool mounting bar. A flat forming tool can be fixed to this, as shown in FIGURE 34; and by using the operating lever (FIGURE 35), the tool is brought into contact with the lower side of the rotating tubing, which, softened in a flame, is pushed into shape. A socket for a cone-and-socket joint is easily made by setting up the flat forming tool at the proper angle. The tool mounting bar is fitted with an adjustable mechanical stop which

GENERAL DESCRIPTION

is also an electrical contact for giving visual indication on a lamp when pre-set limits are reached. This arrangement is useful for repetition work. Generally, when constructing specialized apparatus for research work, operations with tools are performed by holding them in the tailstock chuck (not rotating), or in the hand tool holder (FIGURE 34), which is held in the hand.

The mandrel bore size is 32 mm, so that tubing of this size will pass right through the headstock and tailstock. The maximum jaw opening is 63 mm with the jaws in their normal position; but they are

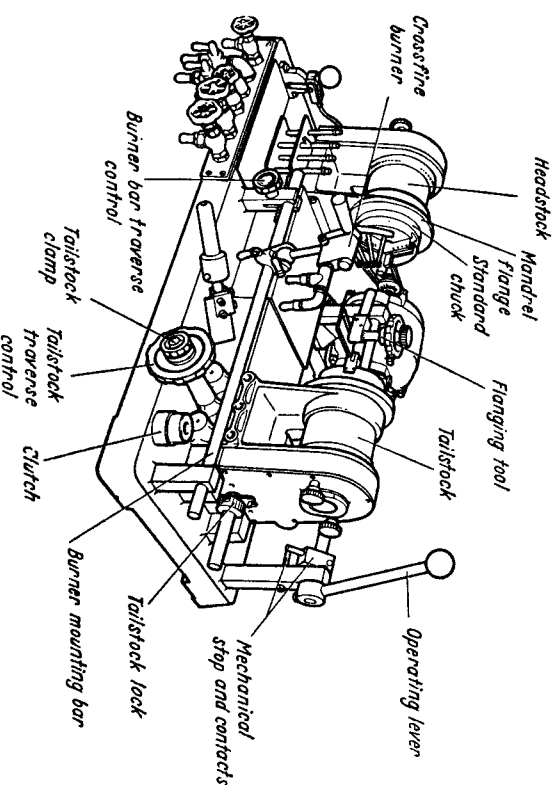


FIGURE 35. Another view of the glass-working machine

reversible, and by reversing them tubing up to 115 mm diameter may be held.

There are various ways of blowing into the rotating work. If the tube, or a portion of it, is long enough and less than 32 mm diameter it may be passed right through the headstock and blown into using a glass-blower's swivel (p. 42). Otherwise spindles or rubber stoppers carrying glass tubes may be used in conjunction with a swivel. The machine itself can be used to put a short spindle in a tube, using, in principle, the method described on p. 56. A blowing fixture can also be fitted to the headstock or tailstock mandrel (see FIGURE 34); this

SOME OPERATIONS WITH A GLASS-WORKING MACHINE

remains stationary while the work rotates, and so it is only necessary to prepare a sufficiently air-tight joint between the tubing to be used and the mandrel with the help of tape or rubber stoppers, or stoppers and tape together.

Sketch of Some Possible Operations

A tube held in the headstock chuck can be flared by any of the methods previously described (p. 72). Flares can be made very easily with a cylindrical carbon rod attached to a special holder fitting into the tailstock chuck, which is not rotated during the operation. The flare is made by pushing the carbon rod, held at the desired angle, into the end of the rotating and heated glass tube, for which purpose the tailstock traverse is used. With this method the end of the tube can be opened out until it resembles a funnel.

To put a spindle in a tube (p. 57), both ends are held in the rotating chucks. It may be desirable to straighten the tube, by heating it in a gas-air flame as it rotates, if it is so bent that it is not held well in both chucks. A spindle is put into the tube by the method described on p. 56, except that the operator's hands are replaced by the two chucks. The length of the spindle which can be obtained is limited by the length of the tailstock traverse.

A tube with a spindle in it can be worked to a tube with a round end by removing the spindle just beyond the shoulder (see p. 57) with a small flame, and then shrinking down and blowing out the end. The blowing can be done while the glass is in the flame. A hole can be blown in the end of the closed tube by heating the extreme end in a small flame. The hole can then be opened out with a tool, using an ordinary flaring method. Two tubes with holes of the same size in their ends can be joined very easily by bringing their hot ends together as they are both rotated, one tube being in each chuck. The joint is finished as described on p. 61. To join big tubes it may be convenient to grind the two ends flat; no holes need then be blown, since the tubes will meet well enough.

To blow a bulb in the middle of a tube, it is held in both chucks; glass is accumulated by pushing the tailstock to the headstock as the glass rotates, and the tube is blown into gently at the same time. The bulb is blown as the glass accumulates; an elegant bulb can be obtained very easily.

An internal seal can be made by Method 1 of p. 73. The hole in the outer tube and the ridge or bulge on the inner tube are both put in on the machine. The ridge can be put in by heating a small portion of the tube as it rotates in both chucks, and quickly pushing one end towards the other with the tailstock traverse. Alternatively, a small

SKETCH OF SOME POSSIBLE OPERATIONS

bulge can be blown until it is the required size. This is conveniently done with the help of a calliper gauge. When making the final internal seal, the inner tube of the seal can be supported in the outer tube with corrugated cardboard. An internal seal can also be made by Method 4 of p. 76. The correct size flange or flare is put on the inner tube with a tool, and a calliper gauge is again used to check the size of the hot glass.

A *Dewar* seal (p. 122) can be made without blowing by flaring the smaller tube outwards until it will nicely meet the larger tube (as in FIGURE 50, *III*). The smaller tube is mounted in the larger tube with the help of corrugated cardboard, and the final joint is made by pushing the flare of the inner tube on to the outer tube with a tool as the glass rotates in the flame. If the tubes are first ground flat at the ends to be sealed, an elegant joint can be obtained very easily.

Graded seals must be used for joining together glasses of such different thermal expansions that a direct seal cannot be made (p. 25). The best way of making a graded seal with the machine is to use the various intermediate glasses as rod. The first piece of glass tubing is held in the headstock chuck, and the first intermediate glass is held in the tailstock chuck; both are rotated. The end of the tube is closed with the rod, which is then drawn off just beyond the closure. The closed tube remaining then has its end worked to a rounded shape. The next glass is attached as a globule on the extremity of the hemispherical end and is worked to another hemispherical end; the process is repeated with the next intermediate glass, and so on. The final tube is joined as a tube.

A graded seal can also be made by fusing together rings of the intermediate glasses, or by fusing on each intermediate glass in the form of tubing, then drawing a spindle, making a round end, and blowing a hole.

The making of glass bellows is an operation which is very greatly facilitated by the machine. The glass is rotated in a small hot flame and blown out to give a thin protruding portion of glass. More of these thin portions are then introduced. Practice is necessary to make bellows with the required degree of flexibility. The flexibility of course increases with the number of thin portions of glass; the fragility likewise increases.

Tapering can be carried out by the methods described on p. 79. When a precise angle is required the flat forming tool on the tool mounting bar (FIGURE 34) can be used.

Chapter 7

METAL-TO-GLASS SEALS

ALTHOUGH the sealing of noble metals through glass has been known for a great many years, even centuries, the extensive use of metal-to-glass seals in laboratories has developed only in recent years. This recent development has, however, been extremely rapid and a great deal of detailed literature now exists. An excellent collation of much of the data has been made by J. H. PARTRIDGE (1949) in his monograph devoted entirely to glass-to-metal seals. We, in this Chapter, attempt to supply the reader with information and detail sufficient to be useful to the average laboratory experimentalist.

Matched and Mismatched Seals

Where a direct fusion of glass-to-metal or metal oxide is obtained (sometimes termed the "wetting" of metal by glass) two types of seal can result—one matched, the other mismatched, depending on how the relative coefficients of thermal expansion of the glass and the metal compare.

In a *matched seal* the stress is kept within safe limits by close correspondence of the two thermal expansions throughout the temperature range over which the seal has to be worked during its manufacture.

A *mismatched seal* is one in which the glass and metal have differing coefficients of expansion. Here the resultant stress is reduced to within the limits required to avoid fracture by special shaping of the metal at the junction. In some cases relief to a small extent is obtained by the use of a ductile metal which gives slightly, so reducing the stress in the glass.

When it is required to seal a metal into a glass with which its coefficient of thermal expansion is greatly mismatched, graded seals are often employed from the given glass, through one or more intermediate glasses, to a glass which will form a matched seal with the given metal. *Kovar-to-Pyrex* seals (see below) are a good example of this type.

A further combined characteristic that a glass and a metal must have before they will form a seal, over and above that of the provision

MATCHED AND MISMATCHED SEALS

by some property or other for the relief of stress in the glass, is that the glass must wet and adhere to the metal or its oxide.

Another important consideration that must be given when employing a matched seal in the laboratory is the temperature range to be encountered during use of the seal. The absence of stress relies on the matching of thermal expansions throughout the temperature range encountered when the seal is worked, and this matching must continue over the temperature range in which the seal is to be used, else fracture may result. This factor must be watched when, for instance, the metal-to-glass seal is to be immersed in, or is near to, refrigerants.

Sealing Tungsten into Pyrex

Pyrex (Corning No. 7740) adheres well to tungsten oxides and will give a vacuum-tight seal with this metal provided the joint is made carefully. The seal is mismatched and an upper limit to the diameter of tungsten rod that will give a satisfactory joint is 0.5–1 mm. The exact figure varies; the most important factor is probably the severity of the service requirements.

For vacuum work tungsten rod is superior to wire because its method of manufacture leaves the metal with less surface cavities and less fine channels running within it. Centreless-ground tungsten rod is advised. The rod is first beaded, and for this a small length of well-fitting, thin-walled, *Pyrex* capillary is drawn; a piece about 1 cm long and approximately 0.5 mm wall thickness is required. The portion of the tungsten rod to be beaded is cleaned by heating to dull red heat and bringing it immediately into contact with crystalline potassium or sodium nitrite, either in the form of a stick or as a small heap of powder balanced on a spatula. A strongly exothermic reaction occurs during which the metal continues to glow red hot and is rapidly dissolved. The reaction is allowed to proceed for a few seconds, then the rod is washed free of the nitrite and wiped dry. The surface of the metal is left clean. It is then reoxidized slightly in a just non-luminous flame until it appears coloured a second order green or blue.

The piece of glass capillary is immediately slipped over the prepared portion of the rod (FIGURE 36, *I*) and heated progressively from one end to the other. The rod must not be allowed to cool before this is done or the oxide layer may break away from the surface of the metal due to differing thermal expansion coefficients. As the glass shrinks on to the metal and adheres to it the air is slowly driven out ahead of the shrinking portion (FIGURE 36, *II*). The metal, when in contact with the glass, appears larger in diameter and assumes a reddish colour so that the progress of the beading is easily observed.

METAL-TO-GLASS SEALS

The finished bead, shown in FIGURE 36, *III*, will, when made correctly, be coloured something between a straw yellow and a reddish tan. If desired, the bead may be enlarged at one end or in the middle to assist in fusing it into a small bore tube as indicated in FIGURE 36,

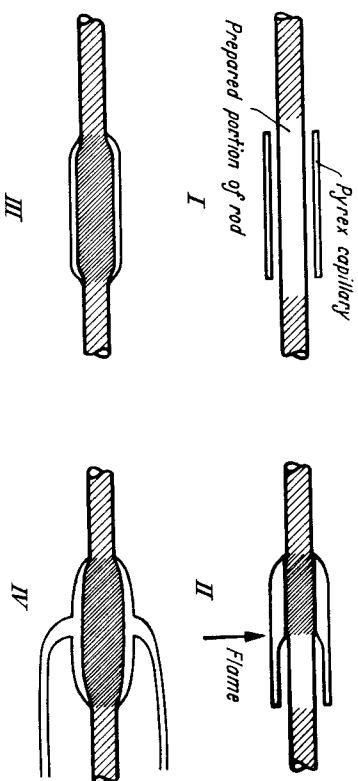


FIGURE 36. Tungsten-to-Pyrex seals

IV. A hole is made at the end of the tube, just large enough for the bead to fit closely, and the end of the tube is fused on to the bead with a hot pin-point flame. The resultant seal should be annealed well and is usually a shade darker in colour than the initial bead. If the tungsten wire or rod has to be cut to length, this is best done by grinding or dissolving in molten potassium or sodium nitrite. Any attempt to cut tungsten results in the metal being torn and in the opening of fine channels which may cause leaks.

Wire Holders

The tungsten rod during the operations described above can, when of sufficient length, be held in the fingers; but with short pieces of rod

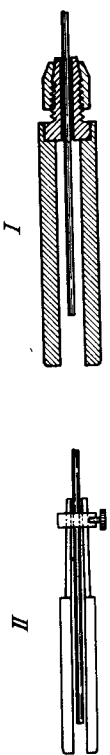


FIGURE 37. Wire holders

some method of gripping is necessary. An ordinary pair of workshop pliers will suffice, but if the metal-to-glass seals are made frequently it is convenient to have at hand some form of small wire holder. A useful tool is seen in FIGURE 37, *I*. A small metal collet head, for

SEALING TUNGSTEN INTO PYREX

gripping the wire, is set into a hollow handle—preferably of wood. Another type of holder, simpler to make, is shown in FIGURE 37, *II*. With wire holders such as these a very even rotation of the wire during beading is possible.

Sealing Tungsten into Special Glasses

With tungsten rods of diameter 0.5–1 mm it is desirable to use special sealing glasses for vacuum work, while with rods larger than 1 mm diameter this becomes a necessity. *Nonex* (Corning 7720), uranium glass (Corning 3320), *G.E.C. W.1*, *B.T.H. C9*, *Dial 36* and *B.T.H. C14*, all form a more matched seal with tungsten than does Pyrex (Corning 7740). *Phoenix* glass, which has an expansion coefficient almost the same as that of Pyrex to which it fuses excellently, seals more

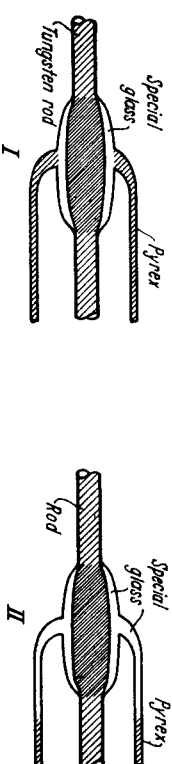


FIGURE 38. Tungsten-to-Pyrex seals with sealing glasses. *Phoenix* can always replace Pyrex in these seals

satisfactorily to tungsten than does Pyrex. It is therefore very useful as an intermediate glass for Pyrex-to-tungsten seals with rod up to about 1.25 mm diameter, and also in making pinch seals.

Of the above special glasses we find that *W.1*, *C9* and *Phoenix* are easily worked and give very satisfactory seals. Very great care is necessary when using *Nonex* to prevent the lead in the glass from reducing with resultant blackening; the tip of an oxidizing flame must be used. The characteristics of sealing glasses are given in Chapter 2. Those mentioned above seal well to Pyrex, though with *Nonex* uranium glass is sometimes used as an intermediate.

When making a seal the tungsten rod is first beaded as with Pyrex. The bead of the special glass may either be fused directly into Pyrex (FIGURE 38, *I*) or first fused into a tube of the same glass and this tube cut near to the seal and fused to a Pyrex tube (FIGURE 38, *II*).

Multiple Wire Seals

When a number of mutually insulated wires have to be carried into a piece of apparatus through one glass tube, a multiple wire seal is necessary. There are several methods for achieving this end, and the one to use depends upon the particular purpose and circumstances.

Multiple Wire Seals when Special Sealing Glasses are not available

One way of making a multiple seal is indicated in Figure 39, *II*. This method is extremely useful when it is required to seal several tungsten rods or wires through *Pyrex*, and special sealing glasses are not available. Each of the wires is first beaded with *Pyrex* as described above. One end of the tube, into which it is desired to seal the wires, is rounded off and the smaller tubes are added one at a time. Tubing of 3-5 mm bore is used for these separate sheaths. Using a hand torch, and employing a fine pin-point flame, it will be found with practice that making a number of small joints, very close together in this manner, is quite straightforward. The central sheath should be added

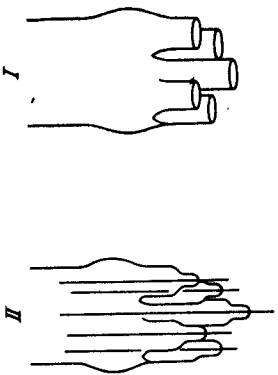


FIGURE 39. Multiple wire seals

first and the remainder positioned around it. As each tube is sealed in position it can be truncated to the required length as shown in Figure 39, *I* and closed with a small expendable cork.

When the multiple sheath is completed each cork is removed in turn and replaced by a beaded rod which is fused into place after careful scraping away of any burnt cork. The finished work should be well annealed. We find the method described here to be very serviceable and useful even when beads of a special sealing glass can be employed.

Pinch Seals

A higher degree of stress exists in pinch seals than when the wires are given separate sheaths. Direct *Pyrex*-to-tungsten pinch seals, therefore, are not satisfactory. However, serviceable multiple pinch seals can be made with most of the special glasses mentioned above. When using *Phoenix* glass a special sealing glass for beading the wires is not necessary.

The end of the *Pyrex* tube that is to carry the wires is heated and shaped with tongs as shown in Figure 40, *I* or *II*. The desired number

of wires, beaded with one or other sealing glass, is held in a clamp of the form shown in Figure 40, *III*, and the latter is supported so that the wires are introduced to the prepared tube with the beads enclosed in the pinched section. The whole of the prepared portion of the tube is then heated strongly and evenly and the glass pinched hard on to the beads with a pair of tongs. The initial beads should be prepared large enough for neighbouring ones to be touching when the wires are clamped in position; this ensures that sufficient of the special sealing glass is present. During the working of the pinch seal all trapped air must be carefully squeezed out. With most glasses flame annealing is adequate.

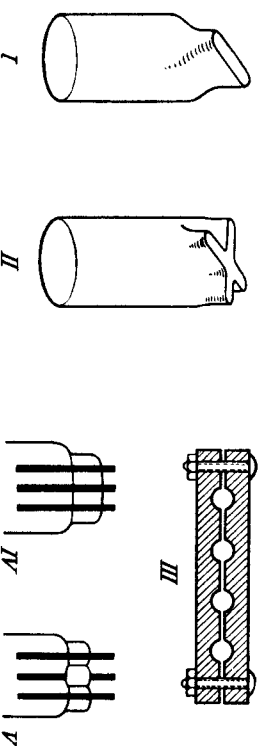


FIGURE 40. Pinch seals

Finished multiple seals of this type are shown in Figure 40, *IV* and *V*. The outer tube may be sealed directly to the apparatus for which it is required, or a *Dewar* seal may first be formed as shown in Figure 50, *VII*, making the multiple seal re-entrant. The re-entrant form is very useful when, for example, the wires are directly attached to, and are supporting, electrodes (see p. 157).

Prepared Copper-Tungsten-Nickel Wires

Long lengths of tungsten rod beyond a metal-to-glass seal are often inconvenient because the metal is so brittle and will not give. Any lateral strain at the end of a long length of rod or wire can easily fracture the metal-to-glass seal. For this reason it is often desirable to use prepared copper-tungsten-nickel wires. These are obtainable, or can be made by butt-welding braided copper at one end of a short piece of tungsten and nickel at the other. Nickel also spot-welds well to tungsten.

These prepared wires are very useful for sealing through *Pyrex* and *Phoenix*. Care must be taken to see that potassium or sodium nitrite does not come into contact with the metal junctions or the tungsten

will not be left clean. When beading the tungsten the glass capillary must be kept short, so that the glass comes only into contact with the tungsten portion and does not overlap either junction, otherwise fracture will result.

Nickel Tipping of Tungsten

In the prepared wires described above, the copper and nickel butt-welds at either end of the tungsten serve to seal up any fine channels that might be present in this metal and to prevent leaks. If the tungsten rod or wire is to be used alone it is sometimes desirable to seal the ends by welding or fusing a coating of nickel over the tip. A fine

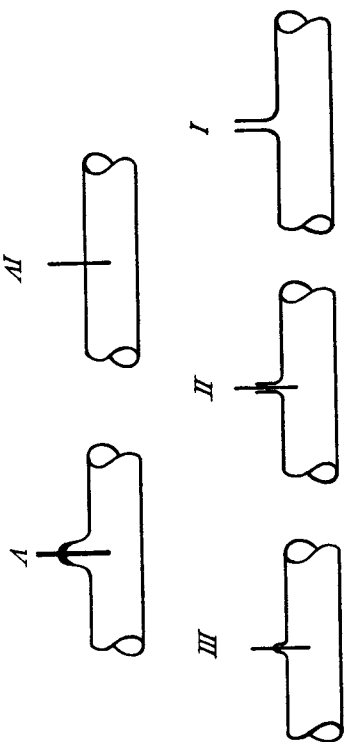


FIGURE 41. Sealing platinum into soda

channel is much more likely in tungsten wire than tungsten rod. With rod the nickel tipping is generally not required.

Sealing Platinum into Soda or Lead Glass

Platinum has a thermal expansion coefficient of $9 \times 10^{-6}/^{\circ}\text{C}$, and therefore seals easily into a number of soft glasses. Fine platinum wire, of diameter less than about 0.25 mm, can be fused directly into soda glass to give a vacuum-tight seal. If wire above that diameter is to be used and the apparatus is required to hold a vacuum, then an intermediate lead glass that will fuse to soft glass should be used; alternatively *Dial 444* may be used as a sealing glass. Platinum oxidizes only slightly and thus no great care is needed when fusing glass to it.

When a wire is to be fused into the wall of a tube without employing an intermediate glass a small tail is first pulled out (FIGURE 41, *I*). This is done by heating with a pin-point flame, touching the tiny hot portion with the tip of a glass rod (or with the platinum wire itself) and drawing out a fine spindle. The spindle is broken off about 2 mm

from the wall and should at this point be of diameter just sufficient to accommodate the wire (FIGURE 41, *II*). The wire is inserted, the fine tail is collapsed and fused on to the platinum to form a small bead *in situ*, which is heated strongly so that it shrinks back into line with the walls of the tube (FIGURE 41, *III* and *IV*). In this manner the seal is made without any appreciable thickening of the glass around the platinum. When the seal is finished the whole circumference of the tube should be heated at this point and well annealed.

Use of an Intermediate Glass

If an intermediate lead or sealing glass is used then the wire is beaded separately and a slightly larger tail is drawn out—sufficient for the whole bead to be inserted. The tail is prepared first so that the hot bead can be inserted and fused into place without cooling. The seal is otherwise finished as above; care being taken not to form a lump in the wall of the tube around the wire. When platinum wire of diameter greater than about 0.5 mm is used it is preferable to add to the larger tube a small T-piece as a sheath for the beaded wire (FIGURE 41, *V*), as described above for tungsten-to-Pyrex seals.

Sealing Platinum into Pyrex

No satisfactory, single, intermediate glass for sealing platinum metal to Pyrex glass is readily available. Very fine platinum wire and very thin-walled platinum tubing can, however, be satisfactorily sealed directly to Pyrex, but such seals are not usually serviceable for vacuum requirements.

Light Platinum Tube

With thin-walled platinum tubing, of wall thickness up to 0.15 mm, the glass is fused around the outside of the tube for a length of 5–15 mm; slight blowing may be necessary. The metal in this form will give sufficiently to take up the stress caused by differing thermal expansions. If used for current carrying, the capacity of the platinum can be increased by inserting a suitable wire and soldering it at one end.

Housekeeper Seal

As described by W. G. HOUSEKEEPER (1923), platinum wire of 0.05 mm diameter can be successfully sealed through Pyrex if the wire is first flattened and the edges filed to a sharp V-shape as shown in FIGURE 42, *I*. The flattened portion of the wire is fused into a tube whose end has first been collapsed down until it will just take the broad part of the wire. Alternatively the wire may be beaded first,

METAL-TO-GLASS SEALS

but in either case a strong heat is required and the glass is pressed into good contact with the metal with tweezers. The entire section of the apparatus around the seal must finally be well annealed. The finished seal is shown in FIGURE 42, *II*.

Use of Mercury Cup

If the platinum wire is to be used as an electrode it may be preferable not to flatten and shape it as in the Housekeeper method, but to use a mercury cup on the outside of the apparatus (FIGURE 42, *III*). The fine wire is fused into a constricted part of a narrow bore tube and the

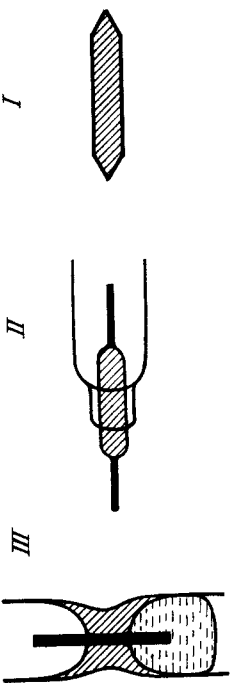


FIGURE 42. Sealing platinum into Pyrex

exposed end, on the outside, is covered with mercury which serves to seal up any leaks that might develop by the glass drawing away from the metal. Another wire dipping into the mercury cup can serve for electrical contact. Clearly this is of no avail for vacuum work, but is useful in making electrolytic cells.

Use of Platinized Glass Rod

A method for effecting a platinum-to-Pyrex seal by platinizing thin glass rod has been reported by G. F. TAYLOR (1929) and described in detail by C. C. COFFIN (1940). Pyrex rod of 1-2 mm diameter is coated with metallic platinum by dipping in a platinizing solution, allowing the excess to drain off, heating uniformly in a smoky flame until a shiny black film appears and then slowly heating to redness to burn the film into the glass (see p. 87). Three to four coats are required and then the platinized rod can be sealed into a Pyrex tube directly. The platinum layer gives electrical conductivity. Coffin reports that 10-15 applications will give a layer sufficient to carry several amps.

Special Alloys for Sealing to Glass

Tungsten and molybdenum, which have been used extensively for metal-to-glass seals, suffer from the disadvantages of being expensive

SPECIAL ALLOYS FOR SEALING TO GLASS

and difficult to fabricate into rod and of being non-ductile. Platinum is not so difficult to fabricate, but it is very expensive. An attempt has been made, therefore, to replace these metals with metal alloys, and extensive research in recent years has been aimed at finding alloys whose thermal expansion characteristics match those of glasses in the range from room temperature to the upper annealing temperature of the glass. The expansion coefficient of glass increases rapidly in the range between the lower and upper annealing temperatures and it has been found that magnetic Kovar-type alloys of nickel, iron and cobalt, sometimes with a little chromium, have thermal expansions which behave in this manner. With alloys of only nickel and iron the increased expansion rate sets in at too low a temperature; however, such wires coated with copper, forming a Dumet alloy, are used extensively even to the present day in the lamp industry for sealing to soft glasses. A detailed survey of all the characteristics of the various alloys used for sealing to glass has been given by PARTRIDGE (1949).

Copper-Clad Nickel-Iron Wires for Sealing to Soda Glass

Seals for carrying small currents through soft glass can be made using Dumet alloy wires. These are drawn from rod which consists of a core of 43 per cent nickel-iron alloy on which about 30 per cent of its weight of copper has been electrolytically deposited. Wires up to 0.5 mm can be satisfactorily sealed into soft glasses. The seals are made in the normal manner but are left not fully annealed so that the glass remains in a state of compression in the longitudinal direction. The deposit of copper is sufficient to raise the total thermal expansion coefficient of the rod, in the radial direction, to the order of that of soda glass, but full annealing would result in a state of tension in the glass along the length of the seal.

If the seals are to be used for vacuum work it is advisable to tip the ends of the drawn wire with nickel in order to seal up any fine channels that may open between the copper layer and the central alloy core.

Nickel-Iron-Cobalt Alloys for Sealing with Intermediate Glasses to Pyrex

A large number of Kovar-type alloys of iron, nickel and cobalt of slightly varying compositions are obtainable and are marketed under different manufacturers' names. Common examples are Ferrico I (54 per cent Fe, 28 per cent Ni, 18 per cent Co), Ferrico II (54 per cent Fe, 31 per cent Ni, 15 per cent Co), Miconel, Telcosel and Nilo. The last three all have a composition of the order 54 per cent Fe,

METAL-TO-GLASS SEALS

29 per cent Ni, 17 per cent Co and expansion coefficient of $4.5 \times 10^{-6}/^{\circ}\text{C}$.

All the above alloys seal well to several special sealing glasses of which *B.T.H. C40*, *G.E.C. Lemington FCN*, *Kodial*, *Corning 7040*, *Corning 7052* and *Corning 7060* are common. The choice of alloy and sealing glass depends (other than upon availability) upon the service requirements. All give very satisfactory seals for normal purposes, the residual strain being slight and negligible; while, for instance, *Fernico II* and *Nicosel* (or *Telcosel*) give practically completely strain-free joints with *Corning 7060* and *G.E.C. FCN* respectively.

For sealing to *Pyrex* glass a further intermediate glass is required in order that expansion coefficients become sufficiently matched. For example, when sealing *B.T.H. C40* or *G.E.C. FCN* to *Pyrex*, *C9* or *W.I.* can be used as intermediates. Another intermediate glass, such as *Dial 43*, may be desirable as a sandwich glass between the *Kovar*-sealing glasses and the tungsten-sealing glasses. A *Kovar* to 7052 or 7040 tubular seal can be joined to *Pyrex* (7740) via *Nonex* (7720) if the tubing is less than 1 cm diameter; while for larger tubing 7052 (or 7040) to 7050 to 7070 to 7740 is necessary.

Metal-to-Glass Seals using Kovar Tubing

Metal-to-glass seals with *Kovar* tubing up to about 8 mm diameter can be made fairly comfortably by hand, but with tubing larger than this it becomes increasingly desirable to use a glass-working lathe so that uniform seals are obtained. Firstly, square edges on the metal must be rounded off, and for this machining or filing is advised rather than grinding, since particles of abrasive are difficult to remove from the metal and great care must be taken. The metal is degreased and then cleaned by dipping in a solution of 10 per cent hydrochloric acid and 10 per cent nitric acid at 70°C , followed by thorough washing and drying. Touching of the cleaned portion of the metal, before the seal is made, should be avoided.

The metal should be outgassed by heating in a stream of hydrogen for 3–4 hours at 900°C , otherwise bubbles will be formed in the glass and an unsatisfactory seal will result. With small size tubing, less than 8 mm diameter, this cleaning in hydrogen is not absolutely necessary but the resulting seal will be less strong hermetically and leaks may develop. A useful makeshift hydrogen furnace consists of a silica tube along which hydrogen is passed and which is kept glowing white hot with a bench torch. The hot hydrogen must not flow into the atmosphere and produce an explosive mixture.

Tubing of the glass to be used should be chosen of diameter such

112

SPECIAL ALLOYS FOR SEALING TO GLASS

that the *Kovar* tubing just slides inside. A short length of cleaned metal tubing is mounted on a glass rod handle with asbestos tape and oxidized by heating to dull red heat. It is allowed to cool slightly and inserted inside the glass tube to a depth of 3 mm. The glass is then fused directly on to the metal starting from one end so that occluded air is driven out. The flame should be played on the glass and not on the metal. The glass tube is finally pressed lightly towards the metal tube so that it builds up slightly over the inside edge of this tube, the flame being directed at this portion during the operation (see *FIGURE 43, I*). Slight blowing may be necessary. The resulting seal is grey in colour and is annealed in the same manner as a glass-to-glass seal. With larger tubing a small 3-mm long ring of glass can be fused on to the metal tube first (*FIGURE 43, II*) and the bulged end of the glass tube sealed to this (*FIGURE 43, III*). Much more satisfactory seals can be made employing a glass-working lathe, for then the glass can be

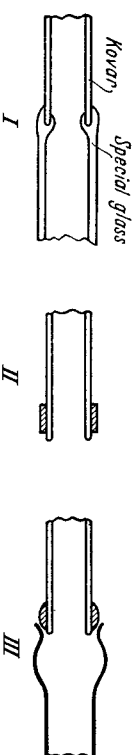


FIGURE 43. Sealing *Kovar* to a special sealing glass

tooled into good contact with the metal and all air carefully squeezed out. Several types of lathe are available; the machine described in Chapter 6 is satisfactory for these seals.

Copper-to-Glass Seals

Most glasses adhere strongly to copper, but the coefficient of expansion of the metal ($17 \times 10^{-6}/^{\circ}\text{C}$) is much higher than that of all known technical glasses, resulting in a greatly mismatched seal. Copper is so ductile, however, that when specially shaped the metal will yield, relieving much of the stress and making the formation of satisfactory copper-to-glass seals possible. The strong adhesion at the glass-metal interface results in a preference for the metal to give rather than to separate from the glass. A large contact surface between the glass and the metal is therefore provided. W. G. HOUSEKEEPER (1923) has reported in detail on the methods for sealing ductile metals to glass with special shaping of the metal at junctions. Oxygen-free high conductivity copper is the best type of copper for seals.

Copper to soft glass seals are most easily made since the difference in expansion rates is least here, but satisfactory seals of copper to *Pyrex* can be formed with a little more care. A further factor with

113

METAL-TO-GLASS SEALS

hard glasses is that a higher working temperature is required and therefore the copper is subject to excess oxidation and the seal to a larger cooling range during which the stresses are set up.

The method for sealing copper wire through any glass is the same as described for platinum through *Pyrex*. However, with copper the metal is very brittle after being worked and the seals are not frequently used. Copper wire can be sealed through glass if it is flattened to 0.05 mm thickness and the edges feathered. The flattened portion is cleaned, pre-oxidized by heating to red heat for a few seconds, allowed to cool, and then given a thin coating of borax. This is effected by dipping the prepared metal in borax solution and drying in the hot air above a flame. Boring of the copper prevents excess oxidation when the seal is worked and is therefore of much greater importance when employing hard glasses. With soft glasses it can be dispensed with but this is not advised if the seal is to be used in vacuum apparatus.

The prepared flattened section is inserted into a glass sleeve shaped to fit, and the glass is fused on to the metal, pinched with tweezers and strongly heated. The seal must be carefully annealed. With soft glasses the copper glass interface is coloured red, while with harder glasses the colour is more of a yellowish red.

Copper-to-Glass Tubular Seals

Tubular seals can be made by several methods reported by HOUSEKEEPER, of which three are described here.

In the first the copper tube is machined so that a cone-shaped flange is formed—30° to the axis of the tube—at the end to be sealed (FIGURE 44, *I*). The metal flange is oxidized and while still hot the molten end of the glass tube is pressed inside the cone and fused into contact. Care must be taken not to let any glass overlap the end of the copper tube, so that a seal is finally formed as shown in FIGURE 44, *II*. Gentle blowing helps to shape the join. That the junction is bulky is the only disadvantage of this type of seal.

In the other two kinds of seal the glass is allowed to overlap the edge of the copper tube, but this edge must be first machined to a fine 5° taper. After pre-oxidizing this feathered edge by heating to red heat for a few seconds a well fitting glass tube can be fused to the outside or to the inside without allowing the oxidized metal to cool. The glass is bulged slightly over the feathered edge thus affording it protection. The seal to use (FIGURE 44, *III* and *IV*) depends upon whether the seal is to undergo further heating or cooling in service, consideration being given to the fact that the copper will attempt to expand or contract much more than the glass.

COPPER-TO-GLASS SEALS

Since tubular seals can be made in this manner to both soft glass and hard glass, an intermediate copper tube can afford a very useful method for making a soda-to-*Pyrex* seal. If *Nonex* is available a stronger join is obtained by using it between the copper and the *Pyrex*, because a lower working temperature is required for *Nonex* and thus, on cooling the worked seal, the copper has a shorter temperature range in which to contract at its greater rate.

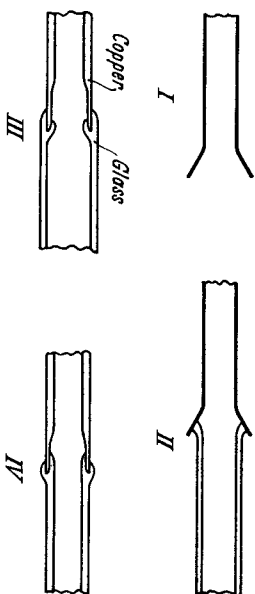


FIGURE 44. Copper-to-glass seals

Disc Seals

Copper disc seals can be used for carrying a wire conductor into a glass tube: two pieces of glass tubing are flared, each at one end, and a copper disc is sealed between the two flared ends. A suitable thickness of disc for tubes of diameter 2 cm is about 0.3 mm.

To make the seal one of the tubes is clamped vertically and the pre-oxidized disc is placed centrally upon the flare of this tube. Working round the tube with a hand torch the glass is softened and the disc fused into place. With the seal still hot, the tube is removed from the clamp and, rotating by hand in a bench flame, the second glass flare

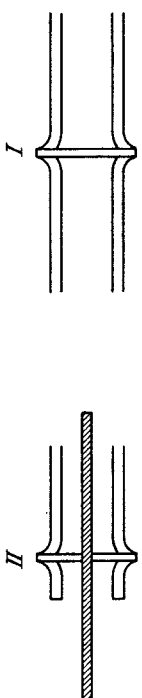


FIGURE 45. Copper-to-glass disc seals

is fused to the disc, centrally and directly opposite the first (FIGURE 45, *I*)—if desired, the lower tube can be left clamped. Great care must be taken during these operations to see that no glass runs over and bridges the edge of the copper disc, otherwise fracture will result. After annealing, one of the glass tubes may be cut off near to the disc and a conductor wire may be soldered through the disc (FIGURE 45, *II*).

METAL-TO-GLASS SEALS

Glass tubing must be sealed on either side of the disc in this manner so that stress is taken up by the disc contracting and not by it bending, which would fracture the seal. HOUSEKEEPER reports that a disc up to 2.5 mm thick has been used but the resulting seal would not stand heating or cooling.

The copper disc, if thin, will distort fairly easily if there is a circular groove in it. This stratagem may be found useful if difficulty is experienced in making a disc seal.

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THE CONSTRUCTION OF SOME TYPICAL SINGLE PIECES OF EQUIPMENT

Chapter 8

In this Chapter we describe the construction of various pieces of apparatus by the methods of glass-working already described. All the apparatus mentioned below can be made by quite simple methods and without the use of the expensive items of glass-working equipment such as lathes, ovens, various wheels and so on. Many pieces of complex-looking apparatus are made merely by a succession of simple operations.

Ampoules

Ampoules (FIGURE 46, *I*) are used when a material, such as a biological fluid, has to be stored in a sealed-off glass vessel. To make an

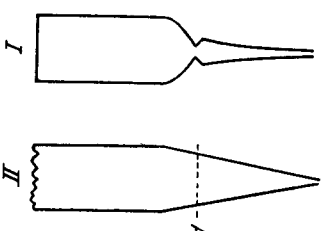


FIGURE 46. Ampoules

ampoule of normal size, a tube about 4 cm long and 1 cm diameter is prepared with a spindle at each end: a uniform taper is made at one end (FIGURE 46, *II*); the tube is then heated at *A* and either rotated against a metal object or else sucked-in so that a shoulder is formed as in FIGURE 46, *I*—this shoulder is not necessary, however, and the operation may be omitted. The spindle at the other end of the tube is

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

taken off and a test-tube end (i.e. a rounded end) is made. This is run down by heating in a flame, and either pressed against a carbon plate or blown out against a carbon plate to form a flat bottom on which the ampoule can stand.

Special glasses for ampoules are mentioned on p. 24; *Monax* is also a good glass for ampoules.

Sealed Tubes for Reactions

Tubes for 'sealed tube' reactions, such as the Carius determination of halogens and sulphur, can be made from *Pyrex*, *Monax* or soda glass. The mechanical strength of the glasses is about the same, but a soda Carius tube is much more likely to crack as a result of thermal strain than a *Pyrex* or *Monax* one. The Carius tubes are usually made from tubing of approximately 20–25 mm diameter and 3 mm wall thickness—*Pyrex* extra heavy tubing of external diameter 22 mm has a wall thickness of 2.5–4 mm and can be used up to 600°C.

To prepare a Carius tube, a length of tubing is cut off slightly longer than the furnace into which it is to go. A handle is attached to one end, which is then drawn off and rounded so that the round end has about the same thickness as the wall of the tube. The tube is cleaned, and the necessary chemical substances are introduced—for a halogen determination silver nitrate is added, using a paper cylinder so that none of the salt adheres to the walls except at the end of the tube. Nitric acid is added with a long thistle funnel, and finally the organic specimen in a small test tube is introduced. The open end of the tube has to be drawn off to a thick spindle and sealed while these substances are in the tube, so that the normal method of holding and rotating a tube horizontally cannot be used.

Two methods for sealing the tube can be used: in one, the tube is held in one hand at about 45° to the vertical, a glass rod handle is attached to the open end, and a thick-walled spindle or capillary is put in, using a bench torch, in such a position that the final sealed capillary will project about 2 cm from the iron tube which will contain the Carius tube in the furnace. Following this operation, the capillary is sealed at the proper place before the tube has cooled. This ensures that the tube is cooling at the moment when it is finally sealed, so that as the gas inside contracts this final sealing is assisted by the atmospheric pressure which pushes the glass in. In the other method the tube is clamped vertically and a handle is attached to the top, open end. A thick-walled spindle is then put in the tube, using a hand torch flame played round the glass; if at first the spindle is too wide and too thin it is slowly thickened by shrinking in a flame which is not too hot. The end of the capillary is sealed off as before.

118

SEALED TUBES FOR REACTIONS

The external diameter of the tube and its wall thickness determine the pressure which it will stand. Failure of a tube is due to the tension at the surface becoming too great. For a cylinder of internal radius r and external radius R , and with an internal pressure in the cylinder of P , the maximum circumferential (or hoop) tension S (see A. MORLEY, 1940) on the inner surface of the cylinder is

$$S = P \left(\frac{R^2 + r^2}{R^2 - r^2} \right)$$

Thus, if the maximum safe tension is 1000 lb/in.² (see p. 10), a Carius tube with 3-mm wall thickness and 2-cm external diameter is suitable for reactions at 340 lb/in.². It is necessary to remember, of course, that the strength of glass under tension is a statistical quantity.

Break-Tip Seals

These seals separate two parts of an apparatus, and they can be broken by a slight mechanical blow. The simplest type is illustrated in FIGURE 47, *II*—the tip is broken off by a piece of iron, usually enclosed in glass, dropping under gravity. This weight is moved and lifted with a magnet.

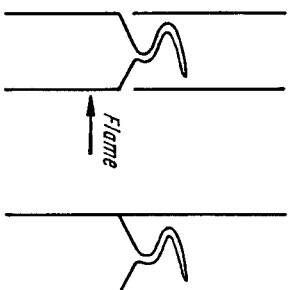


FIGURE 47. Break-tip seals

The seal is made by cutting 1-cm diameter tubing to give one piece with a square end. Near the end of the other piece a spindle is drawn. The shoulder of the spindle is heated in a small hot flame so that it thickens, and then, with a fairly cool flame, the spindle beyond the shoulder is drawn out to a fine capillary about 1 mm diameter. This is heated in a very cool flame and bent first one way and then the other, as in FIGURE 47, *II*. The end is finally sealed off. The tip thus prepared is inserted into the square end of the other piece of tubing, as in FIGURE 47, *I*, and a joint is made by directing a small flame at

119

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

the piece of tubing with the tip, and not at the square-ended piece. The joint is run down and blown out a number of times and then drawn out so that the outer tube has a constant diameter. It is essential to direct the flame only at the tube with the seal on the end of it, because if the other tube is heated strongly it will collapse on to the fine capillary.

Distillation Flasks

A round-bottomed flask can be made into a distillation flask simply by the addition of a side arm, which is added just as a side arm is added to a big flask (p. 92). The angle of the side arm to the neck is usually about 75° (FIGURE 48, *I*).

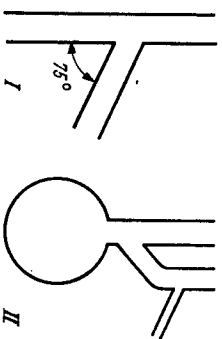


FIGURE 48. Distillation flasks and Claisen flasks

A Claisen flask of the type shown in FIGURE 48, *II*, has a wide side arm sealed into its neck at an angle which, for convenience, is about 45° . Any liquid in the side arm will then readily run back into the flask. A 45° joint of this kind can be made quite easily with a hand torch: a round end is put on the side arm, using a bench torch, and a hole is blown in this round end so that the plane in which the hole lies is at 45° to the tube axis. The wide side arm is joined to the flask neck, and the small side arm is then joined on.

The small side arms are very often broken off distillation flasks, especially in teaching laboratories; the most common place of breakage is near or at the actual T-joint, and the neck of the flask is often cracked. The crack can be repaired easily if the flask is of a borosilicate glass, as described on p. 80, and with great care a similar repair is possible for soda flasks. The remaining stumpy portion of the side arm is then removed with forceps and scissors, the hole is filled in with thin rod, and a new hole is blown for a new side arm. If the remaining part of the old side arm is long enough (over 1 cm)

120

DISTILLATION FLASKS

it is only necessary to put a spindle in it and then blow a hole so that a new portion of tubing can be added.

Distillation Splash Heads

Splash heads are used in distillation to prevent the liquid in the distillation flask from splashing over into the condenser. Several types are used, but probably the simplest is the pear-shaped one shown in FIGURE 49, *VI*. A bulb is blown in the middle of a tube of about 1 cm diameter (FIGURE 49, *I*) and heated at *B* in a large flame, with rotation, so that, by blowing and pulling, the pear-shaped bulb of FIGURE 49, *II*, is produced. The tube at *A* is removed and the top of the bulb is rotated in the flame—this, with gentle blowing, should

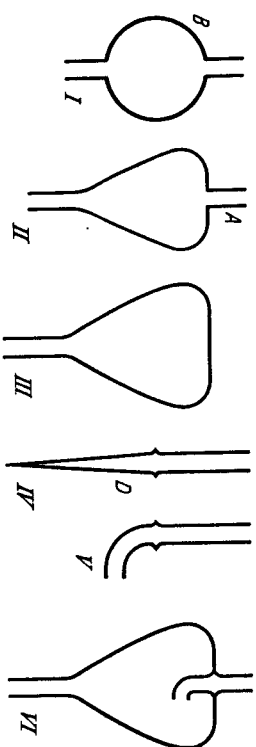


FIGURE 49. The construction of a distillation splash head

produce a fairly flat-topped bulb (FIGURE 49, *III*). The tube to form the top part of the splash head is heated in a large flame and tapered. The top of the taper is next heated strongly with a small flame, and gentle pressure applied at both ends of the tube so that a ridge is formed (FIGURE 49, *IV*). The taper is then heated at *D* in a soft flame and bent round. The curved part is cut off (most easily in the flame) about 1 cm beyond the ridge, and the end is fire-polished (i.e. heated in the flame to smooth it). In the centre of the flat top of the bulb a hole is blown, just smaller than the size of the ridge; into this the curved part is inserted, and an internal seal is made either with a bench torch (Method 1, p. 73), or with a hand torch (p. 76), the former method is preferable. The final splash head is shown in FIGURE 49, *VI*.

Dewar Seals

The Dewar seal is important and useful, although a ring seal can sometimes be substituted in simple apparatus—as, for example, in the cold finger liquid air trap shown in FIGURE 52, *III*. In the making

121

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

of a Dewar seal the preparation of the parts is of extreme importance. Basically, the inside tube is flared at one end, and the end of the outside tube is turned in so that with the two tubes coaxial the inward and outward flares meet as closely as possible.

FIGURE 50, *I*, shows the ideal starting point for obtaining a good Dewar seal. The flare in the inside tube should be just large enough for this tube to be held up by its flared end when sitting inside the clamped outer tube. With small diameter tubing the flange on the inside tube can be made by simply blowing out the end with the glass

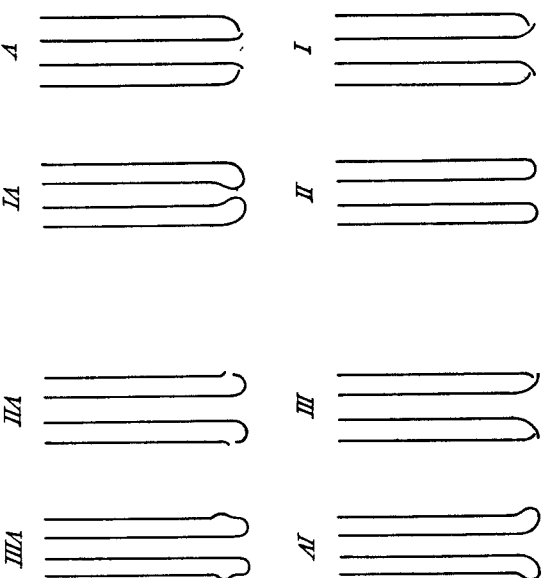


FIGURE 50. Dewar seals

just slightly less hot than normal. To make the Dewar seal using the hand torch, the whole joint is warmed and then a portion at a time worked with a largish flame and enough oxygen to make it just non-luminous. The glass of the inside flare is pushed down with a spike or similar tool on to that of the outside tube, gradually working round the seal, and when blowing becomes possible a pin-point flame is used for blowing out and shaping a portion at a time.

Welding rod is employed when necessary, but if the parts are prepared satisfactorily the flares should provide sufficient glass to alleviate the need for rod. With care it will also be unnecessary to hold the inside tube in position since only part of the joint is heated to

DEWAR SEALS

softening point at a time and, in the final annealing, if the seal is made well the inside tube will tend to hang centrally. However, the inside tube may easily be kept central with asbestos paper or corrugated cardboard or with copper wire that can be dissolved out with nitric acid. In making the seal care must be taken not to collapse the outer tube on to the inner just below the join; otherwise little difficulty should be encountered. The finished seal is shown in FIGURE 50, *II*.

If the starting parts are shaped as in FIGURE 50, *III* or *V*, the resulting seals will be bulged outwards or inwards respectively (FIGURE 50, *IV* and *VII*). These may satisfactorily serve their purpose but require extra careful annealing. If the flare of the inside tube is turned right back on itself (FIGURE 50, *VII*), then the final joint is most easily made.

Dewar Vessels

The work which led to the modern vacuum flask has been described by J. DEWAR (1896). Vacuum jacketed flasks are used extensively in scientific laboratories and can be constructed relatively easily: tubing is chosen to give a flask of the required size and, after the inside tube has been rounded off at one end, a Dewar seal to the outer tube is made at the other end (FIGURE 51, *I*). If a flask with a narrowed neck is required the parts must be prepared as in FIGURE 51, *II*, before the Dewar seal is made.

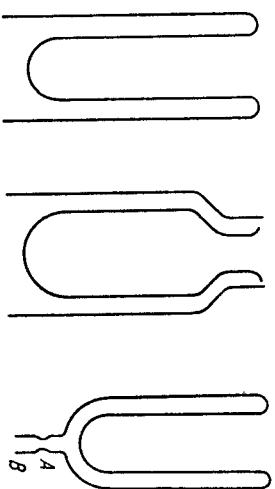
The open end of the outer tube is then drawn off, a 1-cm tube *B* is attached, and a 2-mm internal diameter constriction is formed at *A* as shown in FIGURE 51, *III*. This is best done by rotation in a flame, but, if the tubing is too large for the operator to manipulate, it can be performed with the apparatus clamped as described in Chapter 5. For efficient thermal insulation the inside walls of the jacket should be silvered, and the silvering is performed at this stage of the construction by the methods described on p. 85 and in some detail by W. E. BARR and V. J. ANHORN (1949). FIGURE 51, *IV*, shows a simple stragem for filling and emptying the jacket with the cleaning and silvering solutions respectively. After cleaning and drying, the silvered vessel is ready for evacuation and sealing off.

Evacuation of Dewar Vessels

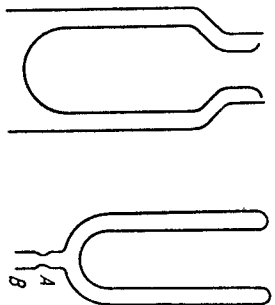
Glass under vacuum slowly releases gas and vapours (p. 16) and therefore, in order that a good vacuum will be retained in the closed jacket, it is necessary to outgas the evacuated jacket before sealing off. As shown by I. LANGMUIR (1918) and described by Barr and Anhorn, the most effective outgassing is obtained when the evacuation is carried out in two or more stages of decreasing temperature.

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

The prepared flask is supported so that the tube *B* can be sealed to a vacuum system. If a suitable furnace is available the flask is surrounded by this, with the constriction *A* still within reach of the hand torch flame; otherwise the heating must be done by flame. A liquid air trap is included between the flask and diffusion pump. With a vacuum of below 10^{-6} mm of mercury the temperature of the flask is raised to near the softening point—the heating may be rapid



I

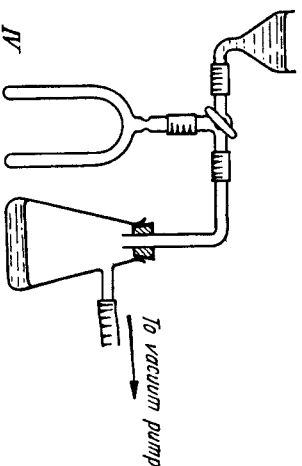


II



III

FIGURE 51.
Construction of
Dewar flasks



IV

with unsilvered vessels, but must be very slow when the walls have been silvered. With *Pyrex* glass (*Corning No. 7740*) a temperature of 550°C is held for about one hour; the temperature is then reduced to about 400°C and this held for a further thirty minutes with continuous pumping. With the flask still at about 400°C the glass is slowly collapsed at the constriction *A*, allowing four or five minutes for this process to enable the heated glass in this region to outgas thoroughly. The tip on the bottom of the flask is rounded and annealed, and the flask then allowed to cool as quickly as possible without setting up strain.

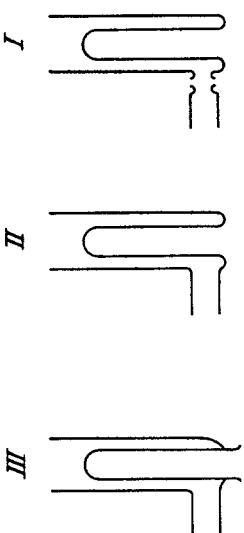
Both commercial and home made Dewar flasks used in the laboratory should be wound along the whole of their length with adhesive

DEWAR VESSELS

tape for safety. This prevents them exploding and shattering in the event of fracture which sometimes results from strain set up during pouring refrigerants, particularly liquefied gases, into or out of the flask.

Cold Finger Refrigerant Traps

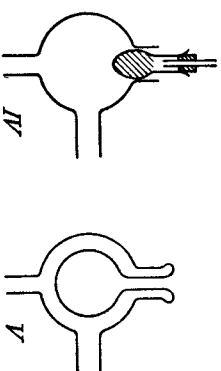
Cold finger refrigerant traps are often used in vacuum lines as a substitute for the more efficient total immersion traps which, however, tend to cut down the pumping speed of the system. In constructing a cold finger trap a Dewar seal is first made, then, before



I

II

III



IV

FIGURE 52. Construction of cold finger refrigerant traps

annealing, a small hole is blown out near to the seal (Figure 52, *I*) and a prepared side arm sealed in position. Care must be taken to keep the far side of the Dewar seal tube hot while the side arm is being added, to prevent cracking; finally the whole is annealed. The finished trap is shown in Figure 52, *II*.

Figure 52, *III*, shows how a quite satisfactory cold finger trap can be made with the Dewar seal replaced by a straightforward internal seal. As before, the side arm must be added immediately the internal seal is completed and the whole annealed together.

For high vacuum work, where it is desirable to shorten the pumping path to a minimum, a useful built-in cold trap is shown in Figure 52, *V*. The outer jacket is prepared from a round-bottomed flask to which the side arms are attached, as shown in Figure 52, *IV*. The inner tube is shaped with a flare at its upper end and its heated

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

Lower end is thickened and collapsed as for making a bulb (p. 71). The lump of white hot molten glass is lowered through the top side arm of the outer jacket and the bulb blown steadily and firmly to the required size. Contrary to expectation it is found that the hot molten glass does not adhere to the cold outer tube when it comes into contact with it, provided the lowering operation is performed quickly. The trap is finished by making a Dewar or ring seal at the upper end. An unskilled worker will find, with practice, that it is possible to make small, bulbular, cold finger traps of this type for pumping lines up to 25–30 mm diameter. Anything larger requires a skilled hand.

With all these cold finger traps the rate of loss of refrigerant is much greater than from the Dewars whose inner surfaces are silvered to cut down radiation losses.

Filter Pumps

It is advisable before making a piece of apparatus such as a filter pump to prepare all the separate components, since the work, if left whilst a component is made, may crack on re-heating—in fact if soda glass is used it probably will do so. Hence for a filter pump of the type shown in Figure 53, *V*, the jet, the cup, the side arm and the tube for the jacket are first made separately.

The jet is made from tubing of about 1 cm diameter: the tubing is strongly heated and a ridge put in. Another ridge is made about 1.5 cm below the first, and a taper is made 4–5 cm long with the shoulder about 1 cm below this last ridge. A ridge is then made at the shoulder of the taper, and the end of the taper is cut off and fire-polished (Figure 53, *I*).

The side arm is made from tubing of about 0.5 cm diameter by drawing a spindle and making a ridge just above the shoulder. The glass is cut off 1–2 cm from this ridge (Figure 53, *II*).

The jet is sealed into the jacket with an ordinary internal seal (Method 1, p. 73), and the side arm is attached at the shoulder; the jet must be perfectly central (Figure 53, *III*).

The cup is then made: a piece of tubing of the same diameter as the jacket (about 1.5 cm) is joined to a tube of the same size as the jet tubing. A slight taper is made in the smaller tubing, and a ridge made at the shoulder of the taper. A bulb is blown at the end of the taper. This is not very easy since the glass is usually very thin at first—the bulb must be blown and shrunk a number of times until enough glass has been accumulated to blow a strong bulb. It should be 3–5 mm diameter (Figure 53, *IV*). The end of the bulb is blown out and spun (p. 73) to give a thistle funnel shape (*B*, Figure 53, *V*). The hole should be just bigger than the end of the jet.

126

FILTER PUMPS

The end of the jacket is rounded off so that when the cup is inserted the jet will be just inside the cup. The cup is next sealed in with an internal seal. The larger tube is then rounded off at *C* (Figure 53, *V*), and a side arm is put on at the bottom and bent round. Lastly, the two side arms are cut off and fire-polished to give the final pump seen in Figure 53, *V*.

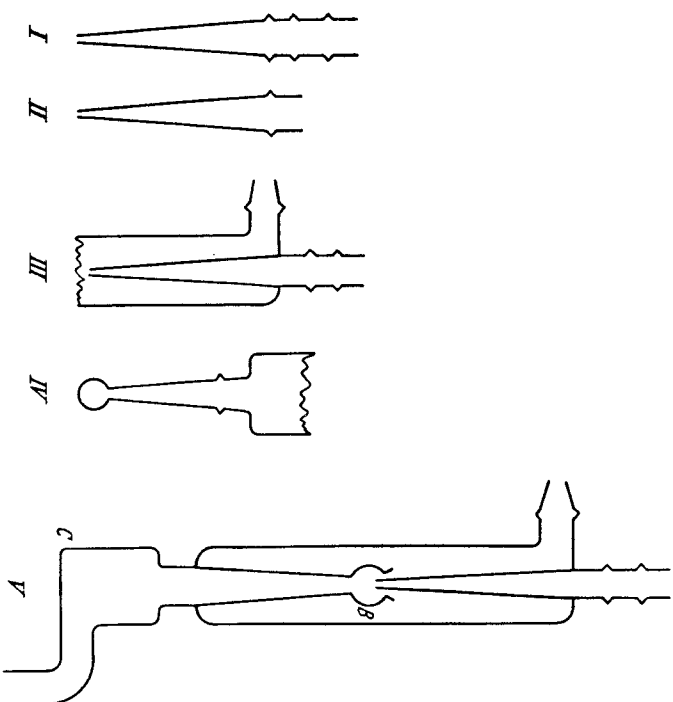


FIG. 53. The construction of a filter pump

Phosphorus Pentoxide Traps

Water vapour and some organic vapours are readily removed from a gas stream by phosphorus pentoxide traps. A simple trap is shown in Figure 54, *I*; *A* is the P_2O_5 boat.

In most cases a tube of 2 cm diameter will suffice for housing the boat, and to one end of such a piece of tubing is sealed a ground glass socket *B*. The other end is rounded off at the required length. At *C* and *D* two T-junctions are formed to tubing 8–10 mm diameter and these side tubes are bent to the desired shape. A stopper *E* is made from the appropriate standard cone as shown. Vacuum grease or

127

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

wax is used for sealing the stopper in place. In this way the P_2O_5 boat can be easily recharged. When building the trap into an apparatus, the side arms at C and D should point upwards so that any phosphoric acid syrup spilt from the boat will not get into the system.

Phosphorus Pentoxide Boats

A satisfactory P_2O_5 boat can be readily made from glass tubing: a short length of suitable diameter is cut along its length (Figure 54, *II*) either on a cutting wheel or by leading two cracks along the length of the tube on opposite sides—this latter operation is fairly easy with soda glass. Using one half, each end in turn is heated and



Figure 54. Phosphorus pentoxide trap

bent upwards, then finally shaped with tweezers to form a closed end as shown. A pin-point flame is used for sealing the seams. Finally the long edges are fire-polished and the boat annealed carefully.

A less elegant P_2O_5 boat can be made much more quickly by the following method: the glass tube to be used is closed at each end with a spindle so that the proper length is obtained. A hole is blown somewhere in the side of the tube, and the glass is strongly heated at one side of this hole and pulled away with forceps. This process is continued until the glass has been removed from one whole side of the tube. The ends are then worked with forceps and tweezers to give the final boat. This method avoids both the use of a cutting wheel, and the cracking of a tube along its length, an operation which does not always go well because the crack may wander in the wrong direction.

Condensers

The construction of a water-cooled condenser is a good example of a case where all the parts must be prepared, and at hand, before the final assembly is begun. There are several methods commonly in practice for setting about the construction; Figure 55 shows one of these.

Lengths of tubing are chosen of suitable diameter for the inner tube and the jacket. An internal seal (ring seal) is formed at one end and a side arm, bulged to take rubber condenser tubing, is immediately

CONDENSERS

added near to the ring seal while the latter is still hot. Thus, after annealing, the stage shown in Figure 55, *II*, is reached. Then, at a point giving the required length of the condenser, the outer tube is collapsed on to the inner tube and the two are pulled off together

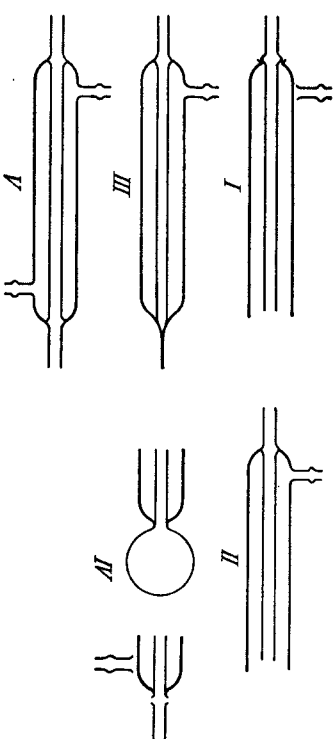


Figure 55. A method of making a condenser

(see p. 76) to a spindle (Figure 55, *III*). Excess glass is pulled away so that a rounded end is formed. Blowing through the side arm at the far end prevents the outer tube shrinking on to the inner behind the ring seal; while blowing down the central tube allows the end of this

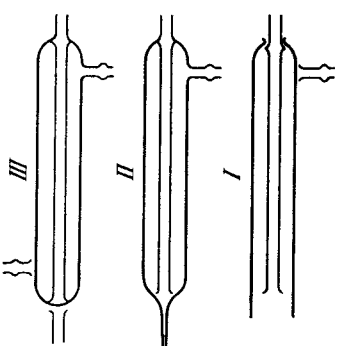


Figure 56. A method of making a condenser

to be thinned and finally blown out as shown. A prepared piece is then immediately added to the central tube (Figure 55, *IV*) followed by the side arm attached to the water jacket. Careful annealing of this whole end is required to relieve as much strain as possible from the apparatus. The finished condenser is shown in Figure 55, *V*. If

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

the outer tube has subsequently to be heated for any further glass blowing, very careful warming is required because, at either ring seal, the outer tube heats more quickly than the inner and therefore tends to draw away from the latter.

A second method of construction is to form the first ring seal and side arm from the prepared parts shown in FIGURE 56, *I*. The inner tube is slightly flared at the far end from the ring seal. In the second stage of the construction the outer tube is drawn off so that the shoulder is very close to this flared end (FIGURE 56, *II*). The shoulder

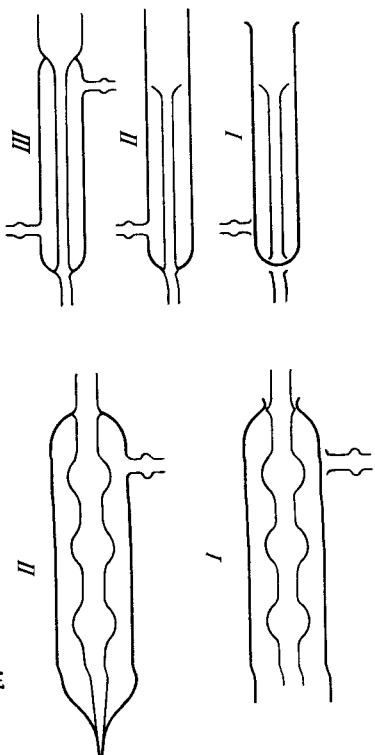


FIGURE 57. A method of making a condenser

FIGURE 58. A condenser with a bulbular inner tube

is shrunk on to the inside flare and, again with blowing through the far side arm, a ring seal is formed. The condenser is then finished as before.

In the third method the outside tube is rounded off at one end and the inside tube prepared with a large flare and a small flare at opposite ends as seen in FIGURE 57, *I*. A butt ring seal is first formed at the closed end of the outer tube. This is blown out as shown in FIGURE 55, *IV*, a piece added to the central tube, and a side arm added to give the stage represented in FIGURE 57, *II*. In the last stage the outer tube is shrunk on to the inner flare forming a large ring seal and the second side arm added. This method gives a finished condenser as shown in FIGURE 57, *III*.

If the internal tube is required to be of some special shape, bulbular or in the form of a spiral for example, then the best method of approach is probably the first method described above: the inner tube is first shaped as required and the outer jacket made to fit as in FIGURE 58, *I*. The ring seal can be formed at one end and the side

CONDENSERS

arm added to give the stage shown in FIGURE 58, *II*. Finally both tubes are drawn off at the far end to give a ring seal and the condenser is finished as in FIGURE 55, *IV* and *V*.

Fractional Distillation Condensers

A condenser of the type shown in FIGURE 59 is often called a fractional distillation condenser and was advocated by H. Jackson, Professor of Chemistry at King's College, London—hence the condenser is sometimes called a King's College condenser. It can be used as an ordinary condenser and as a reflux condenser. To make it, a

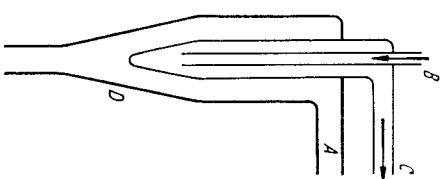


FIGURE 59. A fractional distillation condenser

piece of 3-cm diameter tubing is joined to 8-mm diameter tubing by a taper (*D*). 1.5-cm diameter tubing, for the water circulation, is drawn out to a taper at its end, and closed. 8-mm diameter tubing is prepared for the side arm *A* and 5 mm tubing for the water inlet and outlet tubes *B* and *C*. The 3 cm tubing is rounded off and a hole blown just large enough to let the 1.5 cm tube slide into it. A normal internal seal is next made, but in this case the joint must be blown from both ends. The side arm *A* is then added, followed by the side arm *C*. The larger tubing to which *C* is joined is then rounded off, blowing through *C*, and the central tube *B* is inserted in the usual way, but the internal seal is somewhat more difficult to make since the side arm *C* is already present. The whole apparatus requires careful annealing.

Alternatively, the whole water-circulating part can be made separately and internally sealed into the wide tube, using a hand torch, and

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

blowing from both ends. Another design employs a Dewar seal as in Figure 52, *II*, and the water-carrying tubes are inserted with a rubber stopper.

Fractionating Columns

These are devices for establishing contact between ascending vapour and descending liquid in a distillation. A variety of columns can be made. Perhaps the simplest (Figure 60, *II*) consists of a support in a tube on which a packing rests, made up of short glass tubes, glass rings or pieces of metal gauze. The support for this packing is made

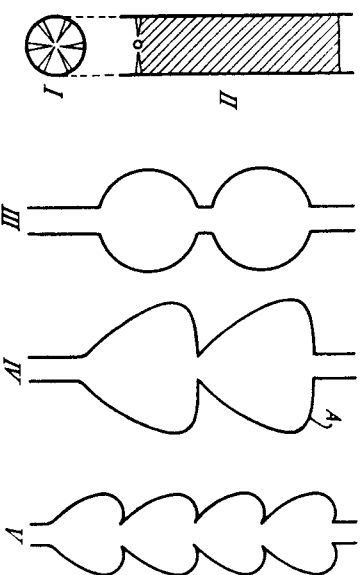


FIGURE 60. Two types of fractionating column

by heating the tube with a small hot flame and pushing the wall in with a spike almost to the centre of the tube; this is done a number of times to give the support shown in Figure 60, *I*. A thorough annealing should be given.

Another fractionating column consists of a number of pear-shaped bulbs (Figure 60, *V*). A series of bulbs is blown (Figure 60, *III*), and their ends are heated and blown and drawn so that pear-shaped bulbs are formed (Figure 60, *IV*). The ends of these bulbs are heated (at *A* etc.) and they are gently pushed in. The column should then be thoroughly annealed. The whole sequence of operations can be carried out very easily with a glass-working machine.

A Fractional Distillation Receiver

A receiver for low pressure distillations designed by G. A. R. KON (1930) is quite easy to make: the joints *D* and *C* (Figure 61) are both standard B.19 joints; the two taps at *A* and *B* are standard designs of three-way taps. For the part *G* a tube of about 3 cm diameter

A FRACTIONAL DISTILLATION RECEIVER

is joined at both ends to tube of about 8 mm diameter. One of the small tubes is joined to the tap *B*. To prevent distortion of the barrel of this tap, the joint should be made 1–2 cm away from the barrel. An internal seal is made into the cone *D* and a side arm attached. The top of the internally sealed tube is then joined to the tap *B*. The two tubes *E* and *F* are joined to the tap *A*. The socket *C* is then joined to the larger tube.

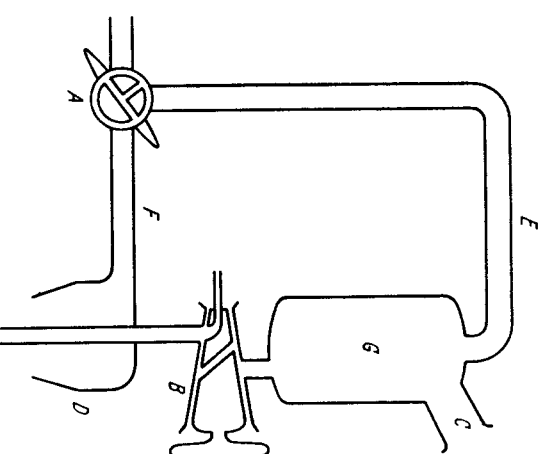


FIGURE 61. A fractional distillation receiver

An advantage of this receiver is that receiving vessels connected at *D* can be taken off and put on without altering the low pressure in the main apparatus. When tap *B* is shut during the operation of changing a receiver the distillate collects in *G*.

Double Surface Condensers

A simple form of double surface condenser that can be made without too great difficulty by an unskilled worker is shown in Figure 62, *IV*. The parts seen in Figure 62, *I*, must first be prepared.

The ring seal at *A* is made in the early stages and is partially annealed. This part of the assembly is then inserted into the prepared tube *B* which is held clamped, and, when sealed well, is itself clamped at *C* (Figure 62, *II*). Using a hand torch and exerting care to warm

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

up the whole area slowly and uniformly the ring seal at *D* is now made by tooling, a little at a time, the glass flare of tube *B* on to the

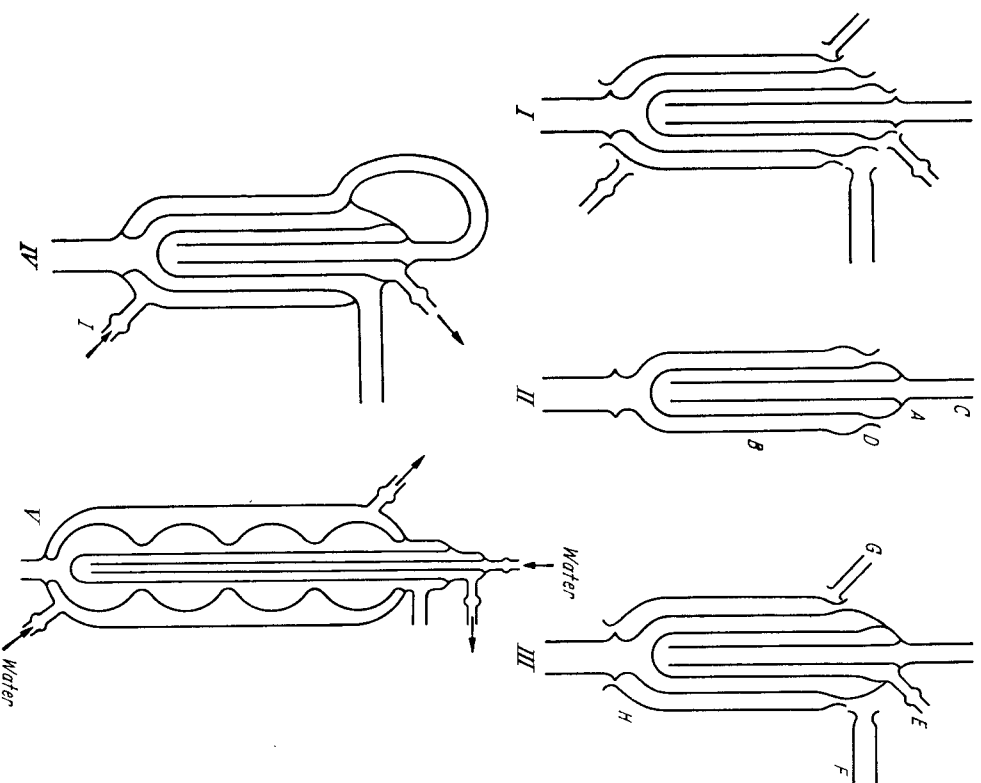


FIGURE 62. A simple form of double surface condenser

bulge of the inner tube. Care must be taken to keep the whole of both ring seals hot while this is being done; a large flame should therefore be used and only reduced to a sharp hot flame for the final blowing out. The water outlet tube *E* is added while the whole area is hot.

DOUBLE SURFACE CONDENSERS

This tube can be held by hand and a pin-point flame used to seal it in position. Every few seconds the flame size should be increased and the far side of the assembly re-warmed before continuing the sealing of *E* in position. Finally the whole section must be very carefully annealed.

The assembly may now be clamped ready for the final large ring seal as in FIGURE 62, *III*. For this operation, and to a lesser extent for that described in the previous paragraph, it is advisable to have a second worker, also with a hand torch, who can keep the far side of the apparatus hot while one side is being worked. Extra careful warming up is required at this stage before the large final ring seal can be made. The warming should be commenced with hot air for about 10 minutes, then with the tips of large luminous flames and finally with slowly increasing oxygen content. It will be found with such a procedure that the initial double ring seal will survive without cracking. The large ring seal is then worked in the same manner as the seal *D*, and, with the entire section kept hot, the side arms *F* and *G* are added. This region with three ring seals needs extremely careful annealing. We have found, however, that flame-annealing is adequate.

The completion of the outer water jacket, with ring seal at *H* and water inlet tube *I*, is relatively straightforward. The tubes *G* and *C* may be joined with a short piece of condenser tubing, though a more elegant finish to the condenser is achieved if they are bent and sealed to give a closed circuit as in FIGURE 62, *IV*. If this is done the upper ring seals must be very carefully protected with asbestos paper, but otherwise the procedure is again straightforward.

The design of this double surface condenser can be changed as desired; for example, FIGURE 62, *V*, shows a double surface reflux condenser which may be made by modifying the above general procedure.

The type of double surface condenser most often found in laboratories is the 'Davies improved double surface condenser' which appeared in 1905, and was designed by J. Davies of A. Gallenkamp & Co. as a direct development from Thresh's modification of the Bidet condenser. All rubber bungs and tubing were eliminated in the Davies condenser, which has retained its popularity for 50 years. To make a condenser of this kind the inner water jacket is connected to the outer one by two internal seals which are made by a slight modification of Method 2 described on p. 75. The rest of the assembly is similar to that for an ordinary Liebig condenser.

Mercury Cut-offs

Mercury cut-offs are used in vacuum apparatus instead of taps when tap grease is undesirable. A frequently used type is shown in FIGURE 63. A two-way tap is attached at the top of a reservoir for mercury.

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

The internal seal at *A* can be made with capillary tube extending right down into the reservoir, but this necessitates very careful annealing. It is easier to first join the capillary to normal tubing of equal external diameter, and then to seal this normal tube into the larger tube. The top part of the cut-off consists of an internal tube at *B* and a side arm. This part is connected to the reservoir by about 70 cm of capillary

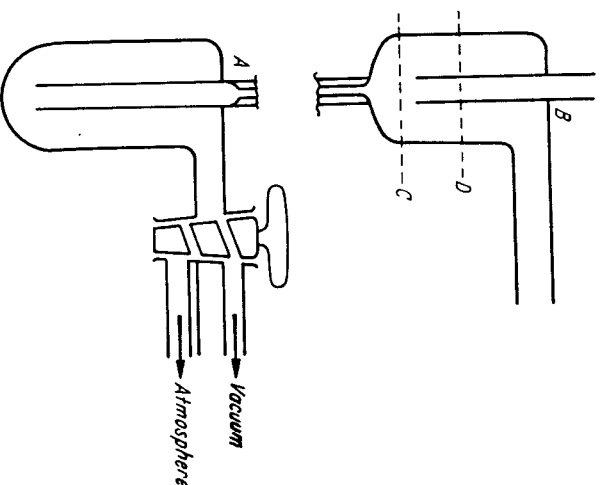


FIGURE 63. A mercury cut-off

tube, which gives strength and reduces the volume of mercury required. To use the cut-off the mercury level is raised and lowered between *C* and *D*.

W. E. BARR and V. J. ANHORN (1949) describe a mercury cut-off with mercury return lines which prevent mercury surging into a vacuum system.

McLeod Gauges

The McLeod gauge is one of the oldest instruments for the measurement of vacua and it has remained virtually unchanged since its introduction in 1874 (H. McLEOD, 1874).

A simple bench type McLeod gauge is shown in FIGURE 64, *III*. Before starting to make such a gauge an estimate should be made of

MCLEOD GAUGES

the volume required to give pressure readings in the required range: a gauge with a volume of 300 ml and using 0.75-mm bore capillary may be used to read pressures of 10^{-5} mm of mercury with reasonable accuracy, but if relatively high pressures are to be measured it will be more satisfactory to use wider bore capillary, say 2 mm. This avoids difficulties encountered with mercury sticking in the capillary. The bore of the capillary should be determined accurately by direct measurement along its length to ensure uniform bore—unless

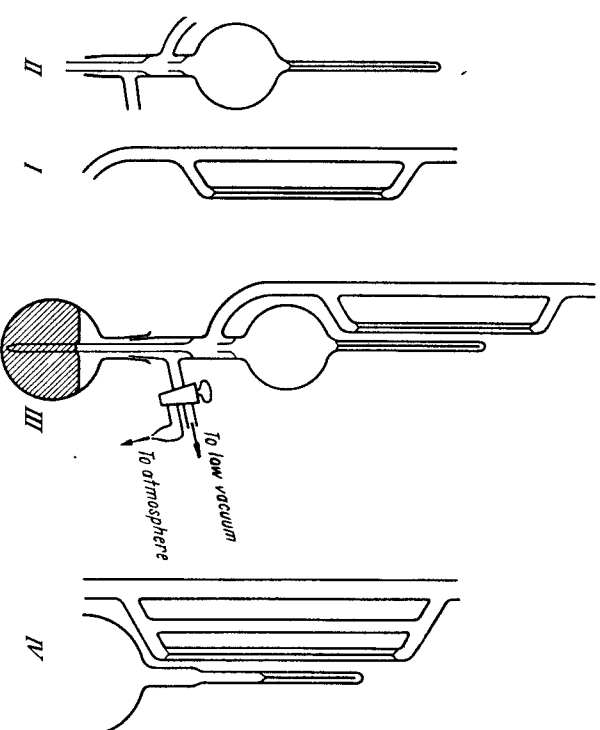


FIGURE 64. A McLeod gauge for bench use

precision-bore tubing is used. This precision-bore tubing may be contaminated by the lubricant used to prevent it sticking to the metal mandrel used in its manufacture, and consequently great care should be taken in cleaning it. Stripping the surface with hydrofluoric acid gives a uniform clean surface if normal cleaning methods fail.

The simplest way of making a McLeod gauge is to make it in two basic sections which are finally joined together. Since the capillary on both parts of the gauge should be identical in bore and in surface properties, it is best to make use of a single length of tubing so that both capillaries are subjected to identical cleaning procedures. The capillary should be carefully inspected for particles of dirt which may

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

not have been removed by the cleaning. FIGURE 64 shows a gauge fitted to the mercury reservoir by a standard ground joint; this is usually the easiest method of assembly.

The part in FIGURE 64, *II*, can be made either by blowing a bulb of suitable volume or by using a round-bottomed flask. If the latter is used a slight bulge should be blown centrally in the bottom so that a sharp shoulder is avoided at the capillary seal. A 15–20 cm length of capillary is sealed to this bulge and its upper end left open. This capillary can then be used as a handle and blowing tube for the subsequent work on the neck of the flask. The neck of the flask should then be drawn down, a smaller tube joined to it and cut off short to form the inner tube defining the cut-off volume. Alternatively, the cut-off volume can be defined by sealing a flanged tube into the neck of the flask, which then does not have to be drawn down. The standard cone is then joined to the mercury supply tube by a ring seal to the flared out end of the tube. This part is then joined to the end of the flask as shown, allowing sufficient room between the two seals for the attachment of the side arm to the vacuum system. The two side arms should be inserted while the glass is still hot from working the internal seals and the whole section annealed thoroughly.

The next step is the calibration of the cut-off volume: this can be done by filling with water and weighing the volume contained in the capillary and bulb up to the cut-off point; for small volumes mercury is a better medium but becomes unmanageable for larger volumes. Once a satisfactory volume calibration has been obtained the end of the capillary is closed to a square end. This can be easily achieved by drawing down a piece of glass rod until it just fits into the bore of the capillary and cutting a short length of this rod to plug the end of the capillary before fusing it. If this is done carefully the capillary will be closed with a square end and the walls of the tube will not be distorted at the end of the gas column, which introduces uncertainties in reading this position. It is important to avoid lens effects here if accurate readings are to be obtained.

The part of the gauge in FIGURE 64, *I*, is simply constructed by joining lengths of normal tubing to either end of the length of capillary, bending these to a suitable angle and blowing out the ends. One end of this composite tube is then blocked with a stopper, and the other end joined to the main tube at an angle by a normal T-joint. A hole is then blown in the main tube for the second T-joint and the shape of the bends in the tubing is adjusted by gentle heating and bending until the end is brought into close contact with the hole. This joint is then completed. The parts *I* and *II* (FIGURE 64) are then joined so that the two capillaries are parallel and a few millimetres apart.

MCLEOD GAUGES

A useful safety precaution is to draw the end of the tube dipping into the mercury reservoir down to a capillary to retard the rate of flow of mercury into the gauge if excessive air pressure is applied accidentally. The drawing down of the air inlet on the two-way stop-cock makes the gauge much easier to operate. Various other methods have been used for raising and lowering the mercury in the gauge but the most convenient and most widely used is that shown—the use of atmospheric pressure and a low vacuum which can be supplied by a normal water pump if necessary. If this method is used it is advisable to have the high vacuum connection at more than atmospheric height above the mercury reservoir.

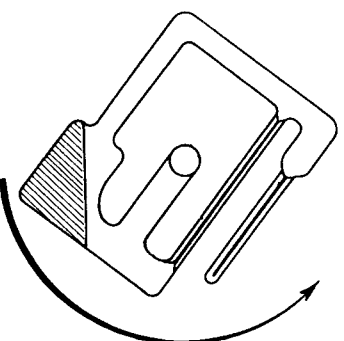


FIGURE 65. A small rotating type of McLeod gauge

The range of a McLeod gauge may be extended by using various sized tubes in conjunction with a small diameter capillary and similarly sized compensating tubes (L. DUNOYER, 1926). In this way comparatively high pressures may be determined on the same gauge as may low pressures. The calibration technique is more complex as the volume is not a simple fraction of the length of the compressed gas column (see FIGURE 64, *IV*).

Small McLeod gauges have been developed for rapid pressure measurements: they avoid the tedious raising and lowering of a large volume of mercury by the gauge rotating about a ground glass joint in the high vacuum connection. A gauge of this type is shown in FIGURE 65. The volumes used are relatively small and can be made conveniently from short lengths of tubing; the reservoir bulb should be large enough to permit free passage of gas into the gauge when all the mercury is in the reservoir. The volume of mercury required is determined by the volume of the bulb and the volume of the tubing

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

—the mercury level in the compensation capillary should be level with the top of the closed capillary when the gauge is in a vertical position and under vacuum. These gauges are best calibrated against a normal type of gauge.

Traps to catch Mercury displaced in Apparatus

When parts of an apparatus contain mercury, as in cut-offs and McLeod gauges, it very often happens that air rushing in or out of the system will displace mercury in an undesirable manner. With cut-offs across a differential pressure, bubbling of gas through mercury is unavoidable, and in tubing of internal diameter less than 3–4 mm bubbling will not occur but the gas will raise a column of mercury in front of it. It is therefore desirable to insert anti-splash traps where-

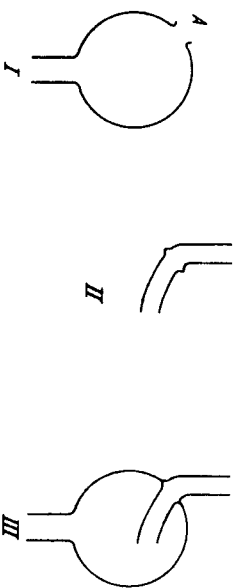


Figure 66. Trap for catching displaced mercury

ever bubbling is likely to occur. A simple form of trap is shown in FIGURE 66, *III*.

In the construction a bulb of the required size is first blown and then a bulge is formed at *A* (FIGURE 66, *I*): the tip of the bulge is blown out resulting in a hole which will just take the size of tubing being used. A short piece of tubing is bulged and bent to shape (FIGURE 66, *II*) and a ring seal is formed at *A*. Care must be taken to see that the lower end of the inside tube finally slopes downwards so that drops of mercury will not lodge in it.

Valves

Valves are of two kinds: those meant to close a system as far as possible, and those designed to operate as check valves which reduce the flow through a system.

A simple and effective check valve which impedes the flow of mercury is seen in FIGURE 67, *I*. It consists of a small plunger floating on mercury, which is pushed into a seating when the mercury level rises. The plunger *P* is made by blowing an elongated bulb at the end of a

VALVES

tube and then drawing out a neck. The bulb is about half filled with mercury and the neck sealed off. The tube *C* is joined to the tube *A* and the edge of the joint at *B* is heated in places and pushed in to form a few spikes to hold the plunger (FIGURE 67, *II*). The plunger is then inserted in *C*; the end of *C* is rounded off and *D* is joined. This must be done with the tubing held vertically, or nearly so, to prevent the plunger coming in contact with the hot glass; the joint can be made very easily with a hand torch.

Valves that are to close a system must have two surfaces which make intimate contact; this is readily effected by grinding the plunger

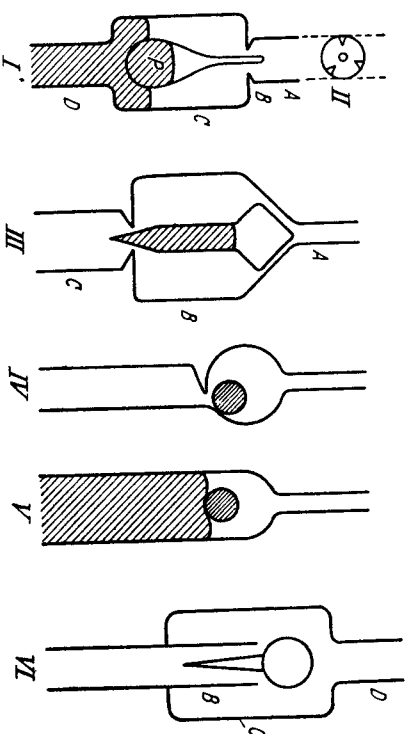


Figure 67. Some kinds of valves

into its seating. A simple design is illustrated in FIGURE 67, *III*: the plunger is made by blowing a bulb on the end of a small tube, heating the end of the bulb, touching it with a piece of hot glass, and pulling the end into a conical shape. The plunger must be able to pass down the tubing to be used as *C*. The tubing *A* is joined to *B* by a taper with an angle equal to that of the head of the plunger, and the plunger is ground into this taper with fine carborundum until a continuous ground band 1–2 mm wide is formed on the glass. The cone is then washed and dried, and *B* is joined to *C*. The end of the plunger is tapered as shown, mercury is added, and the plunger is sealed off and put into *B* through *C*. The junction of *C* and *B* is then pushed in at a few places so that the glass spikes just allow enough freedom of movement of the plunger.

Instead of a glass plunger a ball bearing may be used, as in FIGURE 67, *IV* and *V*. In both of these examples a larger tube is joined to a smaller tube and the joint is ground to fit the ball bearing. The

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

grinding can be carried out with the ball bearing or with a glass grinding tool of diameter equal to that of the ball bearing. This tool is a glass rod with a sphere made on the end by rotating the end in a hot flame. The glass sphere is ground into the joint, and since both surfaces are ground away a slightly tapered seating is produced which is better than that produced by direct grinding with the ball bearing. The ball bearing can float on the mercury as in FIGURE 67, *V*, or be kept in place by a small indentation in the glass (FIGURE 67, *IV*) made by pushing it in with a spike. The former valve is closed by the mercury rising, and the latter may be closed by mercury also, or by a great enough flow of gas or liquid pushing the ball bearing into place. In this case the bulb shown in FIGURE 67, *IV*, may be replaced by a tube just bigger than the ball bearing.

Another form of valve which we find useful is shown in FIGURE 67, *VI*. To make the plunger a sphere is made on the end of a glass rod, and to this sphere a thin glass rod is attached, using a very small flame so that the sphere is not distorted. The sphere is then removed from the larger glass rod with a small and very hot flame, and it is rotated with the thin rod until it is spherical. The glass sphere is next ground into the end of the tube to be used as *B*; this can be done easily by rotating the tubing with a lathe chuck and keeping the sphere stationary. The tube *B* is then internally sealed into *C*, the plunger is put in and finally *C* is rounded off and *D* is added. The tail of the plunger should be long enough to prevent the plunger coming out from the tube *B*. A valve of this kind is useful for liquids and gases; it is opened by pressure of the fluid in *B*, but flow cannot occur from *D* to *B*.

Soxhlet Extraction Apparatus

The apparently complex Soxhlet apparatus (FIGURE 68, *III*) is made by three simple operations: first the large tube *A* is joined to the lower main tube *H*, then the vapour bypass is fitted, and finally the liquid siphon is put on. The main tube is usually made from 3-cm diameter tubing about 15 cm long, and this is joined to 1.5-cm diameter tubing *H*. There is not a continuous tube when these are joined, and hence the joint must be blown from both ends. The joint is made by touching the small tube against the rounded end of the large tube, and then directing the flame against the smaller tube, which is shrunk and blown a few times, and then straightened out. Directing the flame against the smaller tube prevents the larger tube becoming distorted. A hand torch can be used for this joint.

To make the vapour bypass, 1-cm diameter tubing is bent with two right angles, and the ends are cut off so that the tube just fits at *C* and

SOXHLET EXTRACTION APPARATUS

D and is parallel to the larger tube. Holes are then blown at *C* and *D*. One end of the vapour tube is closed with a stopper and the joint *D* is made; then, after removing the stopper, the upper right angle bend of the vapour tube is heated and the tube pushed until the lower end meets the hole at *C*. The end of the vapour tube is then heated and pushed with a spike on to *H* until there are only a few small holes; the joint is finished in the usual way. The top angle is then made smooth and the whole annealed (FIGURE 68, *I*). It is advantageous to have a bunsen burner handy with which the top joint is kept hot while the lower joint is made.

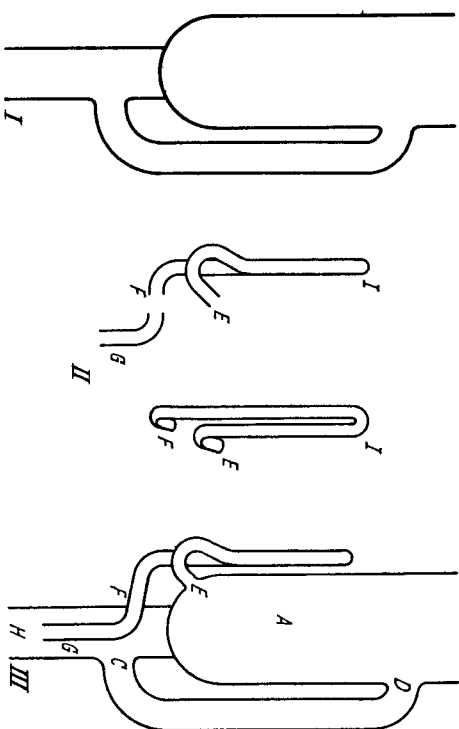


FIGURE 68. A Soxhlet extraction apparatus

For the siphon tube some 2–3-mm diameter tubing has a circular bend made at one end through about 180°, and then about 8 cm from this bend the tube is bent sharply again through 180° (FIGURE 68, *II*). The straight end of the tube is bent about 3 cm below the end of the circular bend, into a right angle in line with the circle. The end is cut off at *F* so that *E* and *F* will meet the larger tubes as shown. Another piece of the small tube is bent into a right angle to make *G*. This piece is internally sealed into *H*, and a fine hot flame is directed against the centre of the joint, which is blown out to a bubble. *G* is made central, and the siphon tube *FHE* is joined to *G* by the appropriate internal seal procedure: *E* is closed with a stopper. This is removed, a hole is blown in the large tube *A* (for joining on *E*) and, with *F* heated, *E* is manoeuvred into position. The joint is then made with the help of a spike as usual. *F* is then made smooth and the whole annealed.

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

In making the second joint *E* it is helpful if the ends of the tube to be pushed to a hole with a spike are thickened and slightly flanged before the first joint is made. This is especially the case for small diameter tubing. The holes blown for both the bypass and the siphon should be so placed that both side tubes are parallel with the main tube.

Mercury Vapour Pumps

Mercury diffusion pumps are standard pieces of laboratory equipment and, for most cases, quite adequate ones can be constructed from glass. In general, Pyrex glass (or similar glass) is necessary because hot mercury vapour circulates and differential strain is

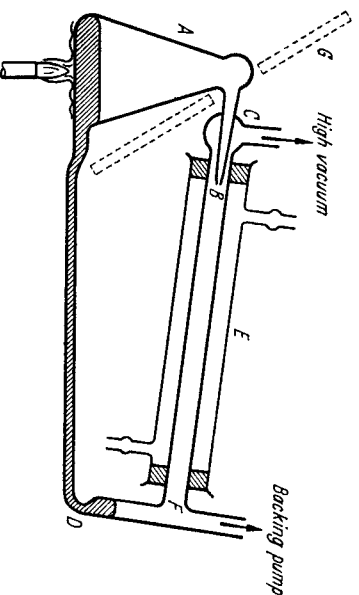


FIGURE 69. A simple diffusion pump

rapidly set up at various, sometimes complicated, joints. Below, however, is described a very simple glass diffusion pump which is quite satisfactory for many purposes, and which can easily be constructed from soda or other glasses.

Very Simple Glass Mercury Diffusion Pump

This pump (FIGURE 69) is due to H. P. WARAN (1923) and is designed such that it may be constructed by the most average of experimentalists. Soda glass is satisfactory because all joints are subject to a minimum, if not a zero, temperature gradient. Pyrex glass would, of course, be an advantage but is not a necessity.

The boiler *A* can be made from an ordinary laboratory conical flask: a side arm is added and then the top of the flask is drawn off and rounded—if a laboratory filter suction flask is available, and has sufficient thermal endurance for a boiler, the side arm already exists and need only be lengthened.

MERCURY VAPOUR PUMPS

The side arm is next bent slightly downwards to form the nozzle *B* which can be tapered as shown if desired. The mouth of the nozzle should be of the order of 5–10 mm diameter, though the size is not critical provided a good rotary, oil, backing pump is to be used. The larger the nozzle the faster the pumping that can be obtained, while the higher the backing pressure employed the finer must be the nozzle and the higher the boiler pressure of mercury vapour.

The ring seal at *C* is made not too far from the flask so that the hot mercury vapour has to travel as short a path as possible. The top of the boiler and the exposed portion of the side arm may be lagged with asbestos rope or paper to reduce loss of heat. The remainder of the pump is easily constructed. The condenser *E* can be any reasonable length, for the length does not affect the pump performance. *D* is a 3–5 mm bore tube for returning mercury to the boiler, and must lie below the level of the mercury in the boiler. The condenser jacket is made separately and is fixed to the inner tube with rubber stoppers or corks which are waxed to give a watertight seal. The jacket must be positioned before the T-junction at *F* is made. If Pyrex glass is employed then the condenser jacket may, if desired, be joined to the inner tube with ring seals at either end in the normal manner.

The boiler can be heated with an ordinary bunsen burner but its base should be coated with a $\frac{1}{16}$ in. layer of asbestos paste—this is absolutely necessary with soda glass. It may be desirable to employ an asbestos board *G* for deflecting radiant heat from the boiler, thereby protecting the ring seal and the waxed cork of the water jacket.

Waran reports that an average pump of this design, working against a backing pressure of less than 1 mm of mercury, will pump a 1 litre volume at atmospheric pressure down to 10^{-5} mm of mercury in 2–3 minutes. When a rotary backing pump is not available, a diffusion pump with a 5 mm nozzle can be operated, with less efficiency, against a backing pressure of 7 mm which can be obtained with an ordinary water vapour pump. Sufficient heat must be supplied to the boiler, in the latter case, to produce a mercury vapour pressure of 1–2 cm.

Glass Mercury Diffusion Pump suitable for High Vacuum Work

A useful glass diffusion pump, due to J. Pollard, is shown in FIGURE 70, *VIII*. This pump is quite satisfactory for high vacuum work, and our measurements on pumps made by hand torch methods show that a pumping speed for air of about 10 litres/second at 10^{-5} mm of mercury can be obtained at the mouth of a pump of this design. FIGURE 71 shows some results for the speed of this pump communicated to us by J. Pollard.

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

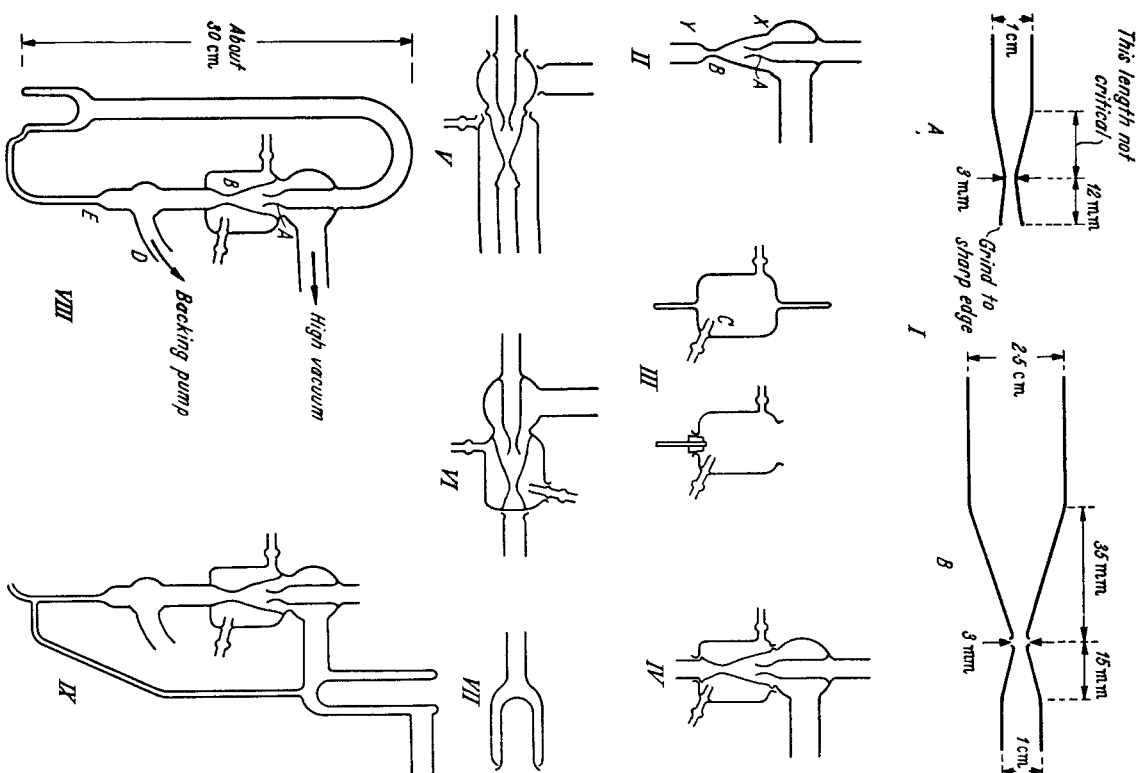


FIGURE 70. A mercury diffusion pump due to J. Pollard

146

MERCURY VAPOUR PUMPS

Two 3-mm internal diameter constrictions at *A* and *B* form the pump 'jets'. These tapered jets are made in the first stages of the construction. The small jet *A* is drawn down from 1 cm tubing by careful rotation and shrinking in a not too hot bench flame. Care is necessary to avoid thickening the glass overmuch. The angle of the taper is not critical but should be approximately that shown in FIGURE 70, *I*. The open end of this jet is ground to a sharp edge.

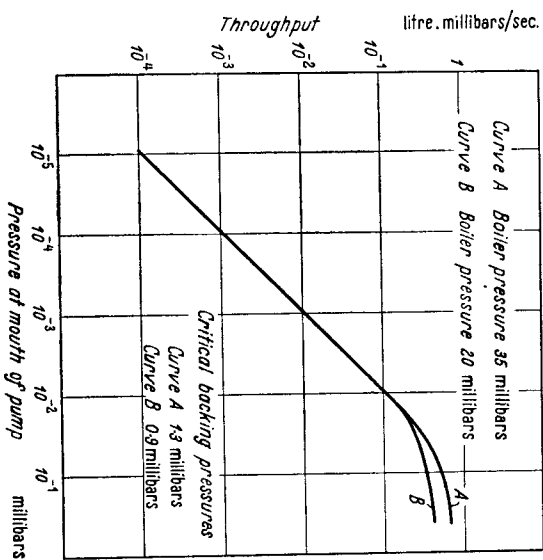


FIGURE 71. The pumping speed of a Pollard type pump

The larger jet *B* is more easily made in two parts. The larger, 25-mm diameter tube is drawn down to the required taper to give a cone-shaped end. As small a hole as possible is blown out at the apex of the cone with a pin-point flame. A 1 cm tube is similarly drawn down and its end blown out to a tiny hole. For joining these two tapers, to give the jet *B*, the tubes are best clamped, since extremely skilful handling is required to avoid buckling if they are rotated by hand. A relatively cool flame will be found adequate for this joint. The joint must be tested for leaks before the pump is assembled (see p. 82).

The larger tubing of the jet *B* is drawn off at the required length, the end enlarged by blowing a slight bulb, FIGURE 70, *II*, and the small jet *A* inserted into place by forming a ring seal as shown. The side arm is added while the seal is still hot and this portion is then annealed. The ground end of *A* should be 35 mm from the constriction in *B*.

1

147

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

Next the water jacket is prepared. A piece of tubing of internal diameter about 40 mm is drawn off to a spindle at both ends to give an intermediate piece of the required length. The two side tubes for in-flow and out-flow of water are added at this stage. The in-flow tube takes the form of a jet *C*, FIGURE 70, *III*, and is inserted by means of a ring seal. At the lower end of the water jacket is blown a small hole to take the 1 cm tube. A rubber stopper carrying glass tube will form a handle at this end for blowing the larger hole needed at the upper end of the jacket. The two jet assembly in FIGURE 70, *II*, is then bulged in the regions *X* and *Y* so that the water jacket fits as shown in FIGURE 70, *IV*. The top upper ring seal can be sealed with plasticine, or wrapped with a long strip of thin rubber sheet, so that blowing is possible for making the lower ring seal. If desired, the upper end of the jacket can be permanently plugged with asbestos paper and sealed with rubber strip or a high melting-point wax. However, by clamping the apparatus, and with careful initial warming, we have found it relatively straightforward to make the large upper ring seal of the water jacket at this stage. We advise a proper seal here because a waxed joint is liable to spring a leak.

A more skilled worker may prefer to do the operations in the order indicated by FIGURE 70, *V* and *VI*. Annealing is then only necessary at two stages: first, when the work of inserting the smaller jet, ring-sealing the upper end of the water jacket and adding the large side arm is completed; and secondly, when the work at the lower end of the water jacket is completed.

When the water jacket is in place, the tubing below the jacket is bulged and the backing pump side arm *D* is added—far enough from the ring seal to cause no strain in the latter. The return drain tube *E* for mercury can now be attached.

The boiler is made separately. The inner tube is chosen to fit the heater to be used, and the outer tube is of such size that the annular space has a volume of 30–40 c.c. to just above the top of the inner thimble. A boiler holding more mercury can also be used.

A piece of outside tubing is first drawn off and attached to a piece of 1.8 cm tube; then the other end of the large tube can be drawn off and blown out at the required length. The inner thimble is shaped separately so that a Dewar seal can be made as shown, FIGURE 70, *VII*. The annular space is usually quite small and care must be taken with this Dewar seal to keep the open end of the thimble from shrinking with the result that the heater cannot be inserted. The side tube for returning mercury is added while the Dewar seal is still hot. The completion of the pump after this stage is straightforward.

The heater can be made from *Nichrome* wire wound in the form of

MERCURY VAPOUR PUMPS

a fine spiral which is itself spiralled round a cylinder of fire-clay. A thin piece of sheet mica serves to separate the glowing wire from the inner glass of the thimble. A 150 watt heater is adequate. The tube from the boiler to the first jet should be lagged along the whole of its length with asbestos rope.

If desired, a cold finger refrigerant trap can be built in, as shown in FIGURE 70, *IX*, with a draining tube for returning condensed mercury to the boiler. The point of return must be below the mercury level in the boiler so that the high vacuum is separated from the backing pressure by mercury in the draining tube. It has been found desirable to have some form of water-cooled condenser, preferably a double surface condenser, in the backing line *D*. This condenser can be positioned conveniently alongside the pump water jacket, and in this way the tube *D* bends upwards so that mercury condensing in the backing line runs back into the drain tube *E*.

Mercury used in diffusion pumps should be thoroughly cleaned by vacuum distilling. The glassware should be flushed with chromic acid, then rinsed and thoroughly dried before filling with mercury. Any grease left inside the glassware will have a most deleterious effect on the performance of the pump. If rubber tubes are used in the backing line these should be freed of sulphur by boiling in concentrated sodium hydroxide solution.

Electrolytic Gas Generator

A very convenient method of preparing small quantities of pure hydrogen is by the electrolysis of a 30 per cent solution of caustic soda (A. FARKAS and H. W. MELVILLE, 1939). A cell for this purpose (FIGURE 72) can be made simply. The cell itself is H-shaped with the top part of the 'H' elongated. The side piece should be of tubing approximately 2.5 cm bore and about 30 cm long, and the cross piece should be of the same size tubing and about 10 cm long. A narrower tube, 1 cm bore, for filling with the electrolyte extends from the mid-point of the cross-piece, and is closed with a rubber stopper when the cell is in use.

Nickel gauze or thin sheet make satisfactory electrodes, which are suspended from tungsten seals in the top of the sides of the 'H'. The gas outlet tubes should be of fairly small diameter (8 or 10 mm), and should include bulbs of about 20 ml volume to allow the gas to bubble through any electrolyte which is displaced into the outlet tubes by uneven gas pressure. Both sides of the gas generator should be connected to an overflow (see FIGURE 72) which allows excess hydrogen and the unwanted oxygen to escape by bubbling through liquid (dibutyl phthalate is convenient). The gas pressures then remain approximately

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

the same on both sides of the cell. We have found that this type of generator is not satisfactory when used on a very small scale.

The glass-blowing of the cell may be carried out in the order considered most convenient by the person making it, but we have found it most convenient to make the 'H' first by making one T-joint in a bench torch and then making the second T-joint with the glass clamped, since this makes it easy to get the two sides of the 'H' the same length. This means that the cell will stand firmly on the bench.

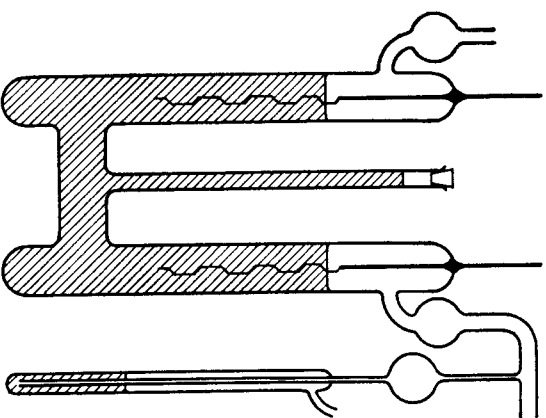


FIGURE 72. An electrolytic gas generator

The filler tube is next attached, followed by the gas outlet tube. The metal-to-glass seals with the electrodes are put in last.

Thermostat Regulators

The regulating device for the heating current of a thermostat may depend on the bending of a bi-metallic strip or on the expansion of a liquid. The latter is more sensitive, and we describe some of these regulators here. An apparatus represented diagrammatically in FIGURE 73, *I*, constitutes the 'make and break' device. When the liquid in the regulator expands, the mercury is pushed up and completes a circuit between E_1 and E_2 , and a relay then operates to cut off the heating current. When the circuit between E_1 and E_2 is broken the heating current is turned on again.

THERMOSTAT REGULATORS

For a sensitive regulator the surface area to volume ratio must be high, so that thermal equilibrium is rapidly approached. Toluene (or other convenient liquid) is used as the regulator liquid, with a mercury column for the electrical contact between E_1 and E_2 . Various designs giving large surface area have been proposed (see J. REILLY and W. N. RAE, 1954). A bulky design is shown in FIGURE 73, *II*; the tap *T* facilitates filling and the adjustment of the regulator for the required temperature. The apparatus is made from 15-mm diameter tubing. The construction is simple: perhaps it is easiest to make four parts separately and join them between *A* and *B*, *C* and *D*, and *E* and *F*.

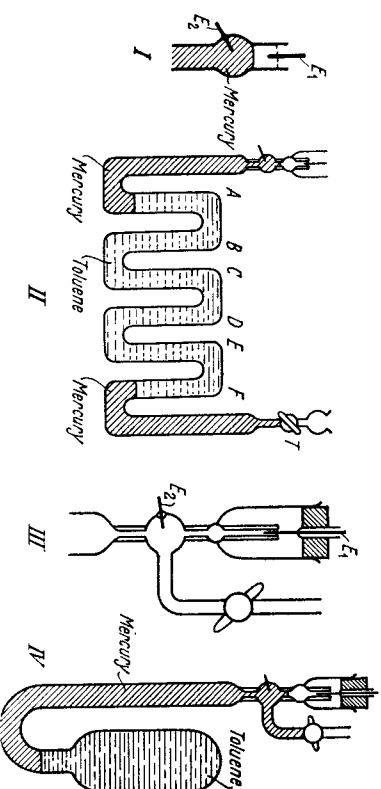


FIGURE 73. Some thermostat regulators

The regulating head is an important part of the apparatus; only a small current should flow, otherwise arcing will occur and the mercury will oxidize. This oxidation is also reduced by keeping the surface area of mercury exposed to the air a minimum by using capillary tube. A useful form of head is shown in FIGURE 73, *III*; the capillary is about 1 mm diameter, the upper bulb about 5 mm diameter, and the lower one about 1 cm diameter. An internal seal is made round the upper bulb; this avoids the thick accumulation of glass which would result from a direct internal seal to the capillary tube. The tungsten rod or wire E_2 of 1 mm diameter can be sealed in without a sealing glass (for a Pyrex apparatus). The electrode E_1 is held by a rubber stopper.

A convenient thermostat control for a beaker is simply a small bulb filled with toluene to which a U-tube full of mercury is attached (FIGURE 73, *IV*). This can be made with 8 mm and 15 mm tubing. The bulb may be filled by successive heating and cooling, with the head in toluene.

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

To keep to a minimum the current flowing through the mercury it is best to employ a valve circuit with a relay, so that the regulator current only controls the grid potential of a valve.

Gas Flow Meters

The usual laboratory gas flow meter is essentially a manometer which measures the pressure difference across a capillary tube through which the gas flows, as in FIGURE 74, *I*, showing a design described by W. BARKAS (1939). This is intended to prevent a sudden change in pressure from forcing the liquid in the manometer into the rest of the

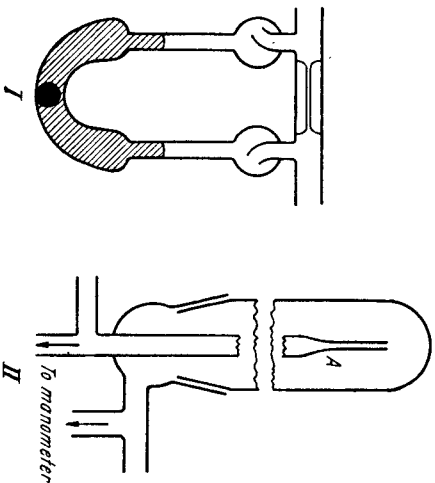


FIGURE 74. Two gas flow meters

apparatus. The bottom tube of the manometer is slightly larger than the rest of the manometer tube, and it contains a fairly closely-fitting ball bearing which is displaced by any sudden flow of liquid and thus blocks the narrower tube. The construction of this apparatus is straightforward. When the capillary tube is joined to the wide tube the join should be as sharp as possible (i.e. there should be a sudden change of internal diameter) if the flow rate is to be related to the pressure difference using an equation for viscous flow of the gas in question; but a direct experimental calibration can also be made, using, for example, a large Berzelius gas holder.

A convenient flow meter is shown in FIGURE 74, *II*. The construction is simple. Splash heads or bulbs should be incorporated in both manometer limbs (as in FIGURE 74, *I*). The capillary tube at *A* may

GAS FLOW METERS

be drawn out as required, in which case calibration is necessary, or a precision-bore tube may be sealed on. With a drawn-out capillary, the correct adjustment is found by breaking off portions of capillary until the proper flow rate is reached; with the ground glass joint arrangement shown this is very simple. A broken thermometer often provides a useful capillary for small gas flows.

Spectrum Tubes

The spectrum excited by an electrical discharge through gas at low pressure can be studied with two simple types of spectrum tube seen in FIGURE 75. These tubes also act as weak sources of certain radiations, such as monochromatic light for an optical spectrometer. The

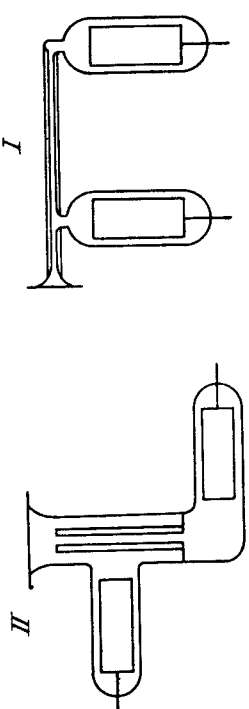


FIGURE 75. Two types of spectrum tube

two tubes shown can be made of Pyrex with fused quartz windows sealed on with *W* wax or a cement (p. 84); but for many purposes these windows are not required. In both *I* and *II* the electrodes are metal cylinders (e.g. aluminium or stainless steel) joined to tungsten rods sealed into the glass. Stainless steel cylinders 1 cm diameter and 3 cm long are suitable—the stainless steel is not so prone to spluttering as most metals. The cylinders are hard-soldered to the tungsten, or bolted on with a suitable connector. If the tungsten rod exceeds 1 mm diameter it should be sealed in with a sealing glass. 1 mm bore capillary is suitable for the tubes.

To fix on the window as shown in FIGURE 75, *I*, a bulb is blown in the capillary, the top of the bulb is blown out to a bubble, knocked off, and the end remaining is fire-polished and finally ground flat. The tube can be filled with gas through a side arm joined to one of the electrode chambers; the spectrum tube may be sealed off, or provided with a stopcock. The capillary tube in FIGURE 75, *II*, is sealed to the glass at one end only of the capillary. To do this, the capillary is joined to a wider tube, which is cut off and flared out to make the

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

necessary flare at the end of the capillary. The remaining operations are simple.

Discharge Tube Lamps

Discharge tube lamps for laboratory work usually consist of three basic parts: the tube to contain the discharge, the electrodes, and a water cooling system. Many different types of lamp have been designed, and while a complete review is outside the scope of this book a few of the more simple ones may be described.

High voltage hydrogen discharge lamps which give continuous radiation in the ultra-violet can be constructed quite simply and elaborate designs are unnecessary for many purposes. A very simple

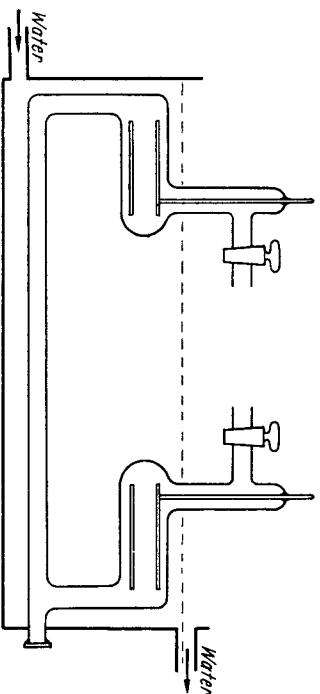


FIGURE 76. A hydrogen discharge lamp

design which should give satisfactory results is shown in FIGURE 76. The lamp consists of a straight tube with a fused quartz or fluorite window waxed to the end and side arms carrying the electrodes and connections for pumping out and filling with purified hydrogen. The lamp is immersed in a trough of water with only the window projecting through a watertight seal in the wall. The electrodes should be of aluminium or stainless steel as these metals are particularly resistant to spluttering under the conditions of the discharge. The tube containing the discharge should be lightly platinized (see p. 87) to facilitate recombination of the atomic hydrogen produced in the discharge.

The chief factors to be borne in mind in the design of these discharge tube lamps are that the electrode material will splutter in the discharge and, unless it is geometrically impossible, will coat the window with a metal film, and that a large amount of energy is consumed by the lamp, much of which is dissipated as heat. This

DISCHARGE TUBE LAMPS

demand an efficient cooling system, particularly in the region of the waxed joint sealing on the fused quartz or fluorite window.

W. WEST (1949) gives a survey of some discharge lamps of different types in which he provides details of the operating conditions for such lamps.

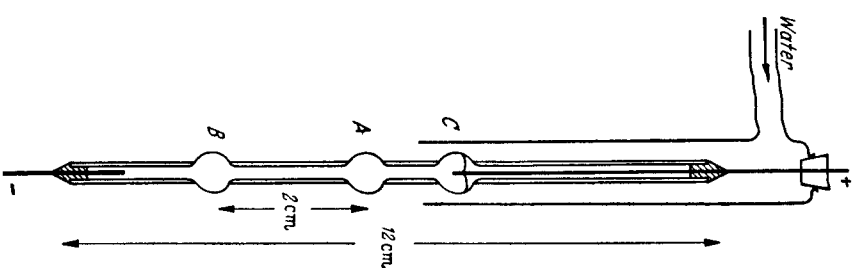


FIGURE 77.
A high pressure
mercury lamp

Air or water cooled mercury discharge lamps find many uses, one of the more obvious of which is the study of photochemical reactions. These lamps are usually made of vitreous silica because of its low thermal expansion, high melting point and its transparency to ultra-violet radiation. Their operating pressure has a profound effect on the spectral distribution of the radiation produced and therefore it is important to consider the requirements in the design of such lamps.

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

E. W. R. STREACIE and N. W. F. PHILLIPS (1938) describe in detail an ingenious design of a low pressure mercury lamp with a rare gas carrier. This lamp was designed for the study of mercury-sensitized photochemical reactions sensitized by the 2537 Å mercury line. The lamp itself in this design is of vitreous silica and Pyrex, joined by graded seals, and mounted in the centre of the reaction vessel, thus giving the maximum amount of reaction possible. Practical constructional details are given by BARR and ANHORN (1949). These authors also give details of the construction of rare gas discharge tubes and display signs.

A simple, high pressure, capillary, mercury arc has been described by F. DANIELS and L. J. HENDR (1932), which, although it has only about twenty four hours useful life, may be useful on account of its very simple construction. The lamp (FIGURE 77) is constructed from silica tubing of 1·2 mm bore and 0·7–1 mm wall thickness. The bulbs *A* and *B* confine the discharge and their spacing governs the operating voltage. The third bulb constitutes the expansion volume and is not filled completely with mercury. The tungsten electrodes are sealed in with de Khotinsky cement (shellac and pine-tar), although it is probable that other stable cements would serve equally well. The lamp is started by heating the column of mercury between the bulbs *A* and *B* with a small bunsen burner flame until the mercury is vaporized. A rapid stream of cooling water is then turned on to prevent the silica tube melting as a result of the discharge. These lamps require a ballasting resistance since they have effectively zero resistance before starting, and 20–500 Ω resistance is generated when operating, depending on the operating pressure and other factors. W. WEST (1949) also describes the operating conditions for this type of lamp.

Apparatus containing many Electrodes

Hand made radio valves and other vacuum tubes containing several electrodes, as for example, mass spectrometer tubes, present special difficulties. In many cases, electrodes, plates and grids can be supported by spot-welding to nickel wires which in turn are spot-welded to tungsten rods that form metal-to-glass seals with the vacuum envelope. Various ways for making the metal-to-glass seals are described in Chapter 7. A pinch seal is often employed, as shown in FIGURE 78, *I*. Careful planning of the position of the final joints, at *A* and *B* for instance, allows for the electrodes to be bent first to any required shape and position. Clearly an infinite number of particular assembly arrangements is possible. Prepared copper-tungsten-nickel wires are often very useful in preparing such electrode assemblies.

APPARATUS CONTAINING MANY ELECTRODES

The above method is not generally suitable when precise spacing of electrodes is necessary. In electron and ion guns, for example, some form of rigid mounting with accurately ground spacings between electrodes is used. The whole mounting is supported in some way in its glass envelope and fine conducting wires make electrical contact between the various plates and their respective tungsten-to-glass seals. FIGURE 78, *II*, shows diagrammatically such an arrangement.

Copper connecting wires are generally used and must be connected to the electrodes before the mounted assembly is introduced into its envelope. The connecting wires can then be drawn one at a time out

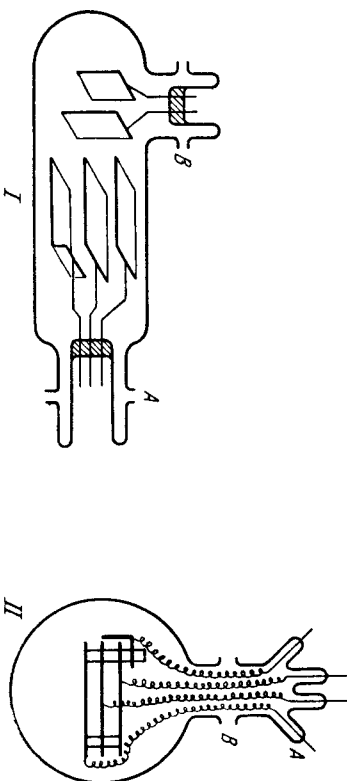


FIGURE 78. Apparatus with many electrodes

through the tube *A*. We have found that fine wires are best (provided low currents are to be passed) because these tend to work-harder, with resultant fracture, less easily than thicker wires. Pieces of thin-walled glass capillary, drawn from 1 cm tubing, make very good insulating sleeves where these are necessary. The wires can be connected with small connectors to the tungsten rods, or welded with a pin-point flame to intermediate nickel wires. This is done as near as is practicable to where the final joint *B* is to be made.

In high vacuum work all metal parts to be mounted inside the glass envelope should be thoroughly cleaned by immersion in appropriate cleaning and passivating dips. Noble metals can be polished with 3/0 or 4/0 (000 or 0000) emery paper. Thorough cleaning of the metals lessens outgassing effects. Any fluxes used in hard or soft soldering must be thoroughly washed away.

To prevent oxidation of electrodes in an envelope which is being glass-worked, a current of carbon dioxide can be passed through the apparatus. This is conveniently obtained from a filter flask containing

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

solid carbon dioxide. A side arm is provided to allow the gas to escape when the final joint has been made in the glass envelope. The carbon dioxide can also be used to blow the final joint; to do this a rubber tube fixed to the side arm is squeezed when blowing is desired.

Leaks for Molecular Flow of Gases

Molecular flow occurs when a gas passes through a tube or a hole with cross-sectional dimensions which are smaller than the mean free path of the gas molecules. The molecules then flow through independently of each other. A leak consisting of a hole in a very thin diaphragm is often useful. For most purposes this can be made of metal foil, but sometimes a glass diaphragm with a hole in it is useful—for example, when free radicals are to be passed through the hole and the diaphragm material must be chemically inactive. We find that microscope cover glasses can be used to make a suitable leak. The cover glasses, about 0.1 mm thick, are first individually measured with a micrometer and sorted out so that about twenty are obtained with very nearly equal thickness. These are covered with paraffin wax on one side, and on each one an area of about 1 mm² of wax is scraped away. They are put in a basin, and the exposed 1 mm² area is etched by placing on it a drop of dilute hydrofluoric acid (20 per cent). All are etched simultaneously and the etching is started as nearly as possible at the same moment for each one. After a time, two or three of the cover glasses will be completely penetrated by the acid, the drop of which will be seen to pass suddenly through the glass. The cover glasses are then all swamped in water, and those which have not been penetrated will have very thin portions of glass about 1 mm square; a thin part can be punctured by a spark so that a smooth circular hole of diameter around 0.01–0.05 mm is formed. A single spark is obtained from an induction coil by giving one light tap to the trembler. Pointed electrodes are used one of which is placed very near the thin part of glass. The cover glass is held in a slot cut in a solid piece of rubber. A spark length of 1.5 cm is suitable, but this can be varied depending on the size of hole required.

Precautions to be taken when using Hydrofluoric Acid

Precautions are necessary when using hydrofluoric acid. It must not come into contact with the skin; the acid can cause severe burns which heal extremely slowly. A large beaker of water should always be handy so that any acid on the skin can be washed off immediately. Running water should also be available. Rubber gloves can be used,

LEAKS FOR MOLECULAR FLOW OF GASES

and they must be free from pin-holes. Barrier creams of the anti-acid type can be used to coat the hands. A paste of magnesium oxide in glycerine is an antidote for hydrofluoric burns.

Bourdon Gauges and Glass Spiral Gauges

Among the methods for measuring gas pressures down to 10^{-1} or 10^{-2} mm of mercury, one of the most convenient is the use of a gauge which depends upon the distortion of a thin glass envelope under varying pressures. These gauges can be used above room temperature, and for corrosive gases which would attack metals.

The simplest form is shown in FIGURE 79, *I*: *T* is a thin-walled glass bulb with one side collapsed inwards. The shape alters with pressure difference across the glass, and the pointer *P* moves. The gauge needs calibration.

The Bourdon tube (as this bulb-shaped part is called) is made by putting a spindle in tubing about 10 mm diameter and blowing a thin-walled bulb just below the shoulder. The bulb is made slightly elliptical by pulling whilst being blown into. The thinner the bulb the more sensitive is the gauge. One side of the bulb is heated by a soft flame until it collapses; the radius of the collapsed side should be slightly less than that of the other side. On gently blowing into the tube the pointer should move several millimetres. The spindle can be left on as a pointer, or be replaced by a thin glass rod. The tip of the pointer should be drawn out to a fine point. The lower internal seal of the gauge can be made with a hand torch, the internal tube being first bulged. A bench torch can also be used. The glass must be blown into from both ends, since the end of the pointer *P* is closed. If the method used for this internal seal involves first rounding off the wide tube and then dropping in the Bourdon tube (Method 2, p. 75), the final hole must be made by pulling away the very hot molten glass with a glass rod, since it is not possible to blow out a bubble of glass.

The movement of the pointer may be observed with respect to a reference pointer which is sealed into the jacket from above. The pointer *P* can also be made to move a glass lever system which magnifies the movement (S. G. FOORD, 1934).

Another type of gauge depends upon the movement of a thin-walled glass spiral—S. G. YORKE (1945, 1948) has given constructional details. The spiral is wound on a carbon mandrel (FIGURE 79, *II*), which is tapered and threaded, and has a hole in it at the top of the taper. The flame is directed against a metal shield just below the mandrel; this gives uniform heating. The thin tubing collapses if heated directly. The thin tubing for the coil is made by drawing out, with blowing, 6-mm diameter tubing to 1.25 mm in a ribbon flame.

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

The thin tube is put through the hole in the mandrel, and 7 or 8 turns are wound. The spiral is cut where it goes into the hole in the mandrel. The gauge can be used with the spiral horizontal (FIGURE 79, *III*) or vertical (FIGURE 79, *IV*). In the vertical form shown a mirror *M* is fixed to the pointer with *Piclen*. A piece of capillary tube *C* is also sealed into the jacket to protect the spiral against excessive movement. An optical window *W* is sealed in or cemented in. A gauge of this type is more sensitive than that shown in FIGURE 79, *III*. Both the Bourdon and spiral gauge can be used with a compensation method or with direct observation of the movement of the

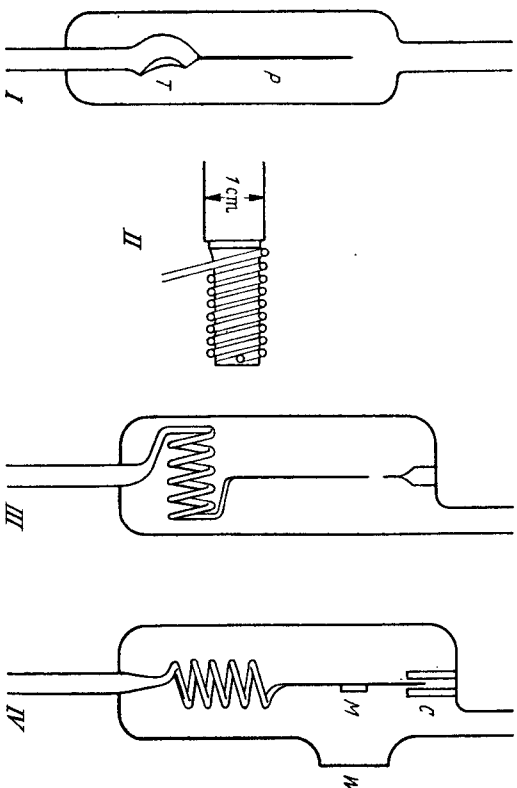


FIGURE 79. Some glass gauges for pressure measurement

pointer. In the compensation method a pressure is applied to the outside of the bulb or spiral until the pointer deflexion becomes zero. For maximum sensitivity it is best to control the gauge temperature by adding an outer jacket through which a liquid at constant temperature is passed.

Circulating Pumps

Simple pumps for circulating gas or liquid can be made using two valves with a chamber between them. One ingenious pump uses a heat engine (A. R. PEARSON and J. S. G. THOMAS, 1925). The hard glass or silica bulb *A* (FIGURE 80, *I*) contains air initially at atmospheric pressure. This is heated with tap *T* closed; and when the

CIRCULATING PUMPS

pressure is 5-8 cm above atmospheric, *T* is quickly opened and closed. The pressure in the bulb suddenly drops and the mercury in the U-shaped tube is set into oscillation and continues to oscillate as long as the bulb is heated. To ensure regular motion the dimensions of the bulb must be properly chosen. Another way of starting the pump is to heat the bulb (to about incipient redness), then to open *T* and to fix on to the outlet from *T* a rubber bulb. By squeezing and

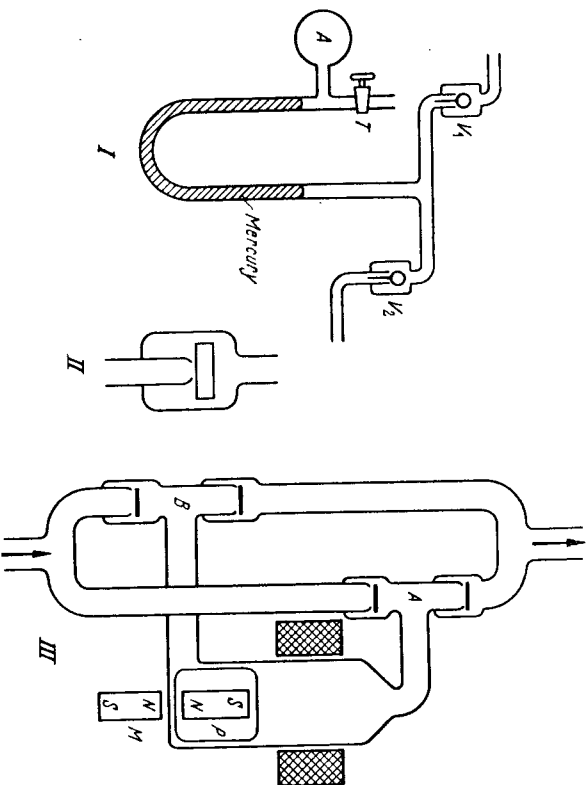


FIGURE 80. Circulating pumps

releasing this bulb the mercury is set into oscillation. The tap *T* is then closed, when the oscillations should continue. The valves *V*₁ and *V*₂ can be of the type shown in FIGURE 67, *VI*. *V*₁ opens and *V*₂ closes as the mercury rises. Another kind of valve is shown in FIGURE 80, *II*. The U-tube can be made of 5 mm tubing and *A* can be a 250 ml flask. A liquid or gas can be pumped.

The pump in FIGURE 80, *I*, pumps at every other stroke, because one stroke is required to fill the chamber between the valves. A. H. COCKETT (1955) describes the pump shown in FIGURE 80, *III*, which pumps at every stroke. The piston *P* consists of a magnet encased in polythene—this avoids trouble which can arise from a glass piston in a glass tube forming powdered glass so that the piston sticks. It is

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

made by pouring molten polythene into a mould containing the magnet—the mould can be a piece of glass tube similar to that of the piston chamber. The piston is finally rubbed with emery paper till a good fit is obtained; it should not be too tight, nor too loose. The piston is lifted with a solenoid of 1000 turns of 22 gauge wire (about 7 Ω resistance). When the current is cut off the piston falls—the magnet *M* prevents it from breaking the apparatus. As the piston moves up, chamber *A* is emptied and *B* is filled; and as it moves down, *B* is emptied and *A* filled, so there is continuous pumping.

Apparatus for Semi-Micro Qualitative Analysis

Most workers develop their own particular methods of performing semi-micro qualitative analyses, and the number of special appara-

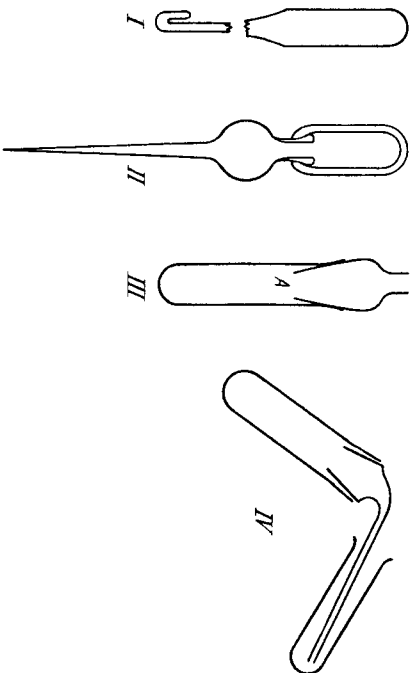


Figure 81. Some devices for semi-micro qualitative analysis

tuses is considerable. Usually, however, certain simple items such as stirrers, spatulas, droppers etc. are used for most of the work. These can be made with great ease. Stirrers and spatulas are made from glass rod: thin rod about 10 cm long and 1.5 mm diameter is drawn out from ordinary rod, or from tubing (p. 58). For a stirrer, about 5 mm of the rod is bent through 180° at the end (Figure 81, *I*). Some thick rod (or tube) can be left on as a handle, as shown. A spatula is made by drawing a spindle in rod, putting a blob of glass on the end and flattening the blob with tweezers. The edges can be bent up with tweezers to form a spoon if desired.

The dropper shown in Figure 81, *II*, has a small bulb so that for maximum uptake liquid does not enter the rubber teat; reagent solutions are then less likely to be contaminated. The capillary tube

APPARATUS FOR SEMI-MICRO QUALITATIVE ANALYSIS

can be drawn after the bulb is blown, and it should not have too thin walls. Some glass should be accumulated before the tube is drawn down.

For detecting gases a tapered tube is made which fits snugly into the top of a small test tube, as in Figure 81, *III*. The hole *A* should be about 2 mm diameter, and the top about 5 mm diameter, so that the reagent can be put in with a dropper. This apparatus is not reliable because the gas usually escapes up the side of the tube instead of through the reagent. The two parts can be ground together to stop this; care is then necessary, if the test tube is heated and cooled, to stop the reagent entering it. A thin tube can be prepared, as in Figure 81, *IV*, for bubbling gas through a reagent: this is again tapered to fit the test tube, and is preferably ground in; alternatively a rubber ring can be used as a gasket. The volume of the gas delivery tube should be small, so the glass tube is first worked till the walls are quite thick by pushing it together while it is rotated in the flame.

Gas Analysis Apparatus

The art of analyzing gases by chemical means has made vast progress since the early researches of R. W. Bunsen. Most gas analysis apparatus, however, still contains a reservoir, a combustion chamber, and an absorption vessel, and formidable-looking apparatus is found on inspection to be capable of assembly by a series of moderately

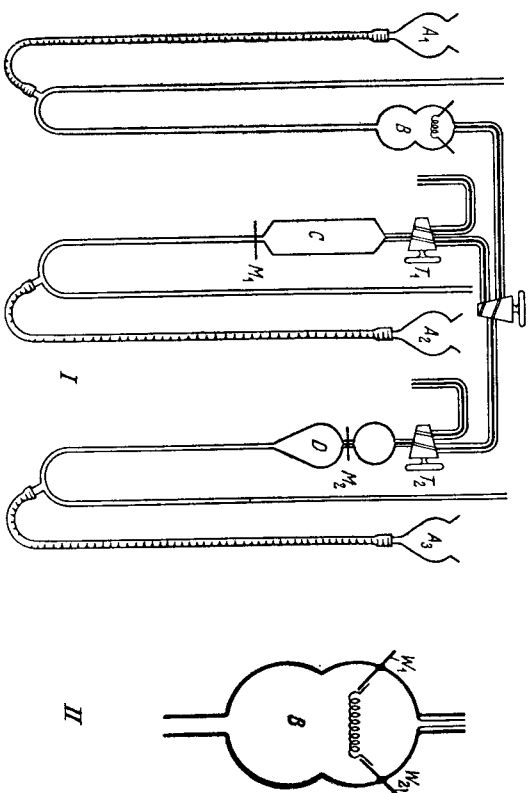


Figure 82. A simple form of gas analysis apparatus

SOME TYPICAL SINGLE PIECES OF EQUIPMENT

straightforward glass-working operations. A simple form of apparatus is shown in FIGURE 82; B is the combustion chamber. The apparatus can be used in different ways: for example, both C and D can be used as reservoirs or absorption vessels— C is usually the reservoir and D the absorption vessel.

The best size for the capillary tubing is 1.5 mm bore; the rest of the tubing in the apparatus is 8 mm diameter. In making the part B , two tungsten rods W_1 and W_2 (FIGURE 82, II) are spot-welded or hand-soldered to a platinum wire coil—if spot-welding is used it is much easier if an intermediate piece of nickel is introduced. The tungsten is beaded, with a sealing glass if desired; two holes are blown in B for the tungsten, which is put in, with the platinum, through the lower tube (FIGURE 82, II). It is convenient if the beaded tungsten just passes through the holes. The volume of the reservoir C from the tap T_1 to the mark M_1 must be measured (if it is needed) before this part is joined to the apparatus. Similarly, the volume between the tap T_2 and the mark M_2 should be found before assembly. A rigid frame is needed for the apparatus because of the great weight of mercury in it.

The mercury reservoirs A_1 , A_2 and A_3 are standard equipment; they can be made by joining at each end about 8 cm of 4-cm diameter tubing to 8 mm tubing. The large portion of tubing is then blown to a bulb of about 7 cm diameter, which is worked by pulling and blowing to a pear-shape. The 8 mm tubing attached to the pointed end of the bulb is corrugated to take rubber tube, and at the other end the bulb is blown out and flanged.

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Chapter 9

THE ASSEMBLY OF COMPLEX APPARATUS

General Observations

THE assembly of complex glass apparatus is comparatively easy provided it is done in a systematic manner. It is most important to work out the general arrangement of the various parts of the system before starting to assemble them, and the sequence of operations should be planned in detail.

The arrangement of apparatus is usually dictated to some extent by the laboratory space available; but within this limitation careful preliminary planning can make the construction, use and maintenance of the apparatus easier. The apparatus should be designed so that all parts are easily accessible and so that its operation does not impose too great a physical strain on the worker due to awkwardly placed components. An apparatus, badly designed in this respect, will probably be broken much more often than a more carefully planned one.

Assuming that the apparatus has been carefully planned and that it is to be a vacuum apparatus, it is generally more convenient to start building it at the pump end of the vacuum line. This allows the joints to be tested before the complete system is finished, which is an advantage if some of the joints are screened by other parts of the apparatus constructed later. Probably the most convenient method of working is to join on each section with its far end finishing in a sealed-off spindle which can easily be blown out preparatory to the joining of the next section. This technique also excludes unnecessary dust and contaminants from the system during building. If some other method of rendering the system air-tight for blowing is used, care should be taken to see that it does not contaminate the glass. Care should also be exercised to avoid introducing flakes of glass into the apparatus when holes are blown in the glass. If such flakes are carried in the air stream during evacuation and lodge in the diffusion pump, its efficiency may be seriously impaired.

The cleanliness of the inside of a vacuum system is of the utmost importance, and therefore care should be taken at all stages to see

GENERAL OBSERVATIONS

that the glass is thoroughly cleaned before assembly and that no unnecessary dirt is introduced during assembly. The introduction of a single length of improperly cleaned tubing can render useless all the effort spent on cleaning the remainder of the apparatus. Open tubes on the apparatus should be blocked in some way, even if not required to be air-tight, to prevent dust from entering them. Sometimes it may be desirable to prevent excessive amounts of moisture from entering the apparatus. For this purpose a tube of drying agent can be included in the blowing tube, although very often large amounts of water vapour enter the apparatus as combustion products of the flame. When glass-blowing is carried out on a system containing greased stopcocks or ground joints they should not be overheated; if they are, grease is spread over the inside of the system. *Silicone* grease on ignition forms a solid mass which cannot be burnt off.

Whatever the problems and however complex the apparatus to be constructed, the sequence of operations should be carried out mentally before any practical work is started. The time spent in planning any piece of practical work is well spent and will probably save time eventually.

Joining Closed Systems

The need for joining closed systems arises quite frequently, and various methods may be used to make the final joint. Difficulty usually arises out of the necessity of blowing two holes in the system in close proximity to one another.

One method of doing this has been described already in the section on McLeod gauges; it involves blocking the first hole with a stopper, blowing the second hole and then bending the glass until the holes are in a suitable position for joining. This method may be varied so that it is unnecessary to allow the glass to cool after blowing the first hole. If care is taken in blowing the first hole, a bubble of glass, sufficiently thin to be knocked off but strong enough to withstand the blowing pressure, can be produced. It is then possible to blow the second hole and remove the thin glass bubbles.

The most difficult part of joining a closed system is making the two prepared tubes meet correctly. This can usually be accomplished by bending either or both of the tubes, and perhaps in addition drawing the tubes out until the desired fit is obtained. Once the joint has been made by pushing the glass together or by filling the gap with rod, the irregularities produced by bending and pulling can be evened out by judicious heating and blowing, although for many purposes they will not have any effect on the efficiency of the apparatus.

Another method which may be useful in joining closed systems is

THE ASSEMBLY OF COMPLEX APPARATUS

described more fully in the next section. This involves making the two final joints simultaneously, at either end of a reasonably long length of tubing, inserted between prepared sites.

Making more than one Joint at once

When several tubes in a complicated piece of apparatus have been cut in order to remove a large or intricate portion for repair or alteration, difficulty is encountered, during reassembly, in obtaining a closed system for blowing. It becomes necessary to make several joints at once and we have found various techniques useful in these circumstances.

Closing Tubes with Ragged Ends—Use of Rubber Caps

The cutting operation when the tubes are not easily accessible usually results in slightly ragged edges. However, the two halves will fit perfectly, and it would be convenient not to have to fire-polish either edge, so that when the reassembly is undertaken the two will still be a perfect fit. If a rubber stopper is used for blocking a cut tube, fire-polishing is necessary in order to make an air-tight seal and in order to reduce the somewhat large danger of splitting the ragged end when the stopper is forced in. Rubber caps are therefore very useful here and, with large tubing, thin pieces of sheet rubber (see p. 35).

Glass-Blowing Two Joints Simultaneously

It is occasionally convenient, both when dealing with complex apparatus and when replacing a short section of a fixed glass tube, to be able to glass-blow two joints at once. This can successfully be undertaken when the two joints are reasonably close together and accessible, and when the tubes involved are not larger than 15 mm diameter.

The interposing portion of glass is first shaped and its ends worked until it fits as closely as possible. It is then clamped in position. By using the hand torch for a few seconds at a time on each joint they can be worked simultaneously, care being taken to heat each evenly all round so that one does not crack in the few seconds that the other is being worked. By using the spike and welding rod where necessary, the two joints can be first sealed and then blown to satisfactory shape. Each joint must be separately and carefully annealed (unless they are so close in the same tube that they can be heated together to the annealing point).

Glass-Blowing more than Two Joints at once

The following method also applies to two joints that cannot be treated by the procedure just described: Each joint is worked in turn

MAKING MORE THAN ONE JOINT AT ONCE

without blowing and sealed up, using the hand torch, welding rod and the spike. Blowing is not possible until the last joint is reached, and so care must be taken not to collapse the tubes too far not to apply too much rod and produce large blobs which will sag. Too hot a flame should be avoided so that the portions to be sealed do not shrink away from each other. Pin-holes will not matter in this part of the operation as long as a pressure sufficient for blowing can be built up. After each joint has been worked in this way, it must be very carefully annealed since the half-finished seal is normally very uneven in thickness and resulting strains can easily cause cracking while the next joint is worked. The joint that will give the most difficulty, either because the tubing is large or because it is inaccessible, is left to last since blowing will then be possible. Finally each joint is carefully warmed up, blown to shape, finished and annealed.

Clearly, when a portion of apparatus has been removed by cutting several tubes, each of the tubes can be drawn off before reassembly and, in the reconstruction, short pieces of new tubing can be used for joining the various pairs of cut tubes in turn. However, when the removed portion can be re-clamped in position with the cut tubes meeting well, it has been found that the above procedure is quicker and more satisfactory, particularly with tubing larger than 15 mm diameter.

Use of Thin Rubber Sheet in making Several Joints at once

With extra large tubing, greater than 35 or 40 mm diameter, a joint sealed with welding rod but not blown out to even thickness is unlikely not to crack on cooling, however well annealed. The procedure in the preceding paragraphs cannot therefore be used when more than one such joint is involved. Thin rubber sheet has been found very useful on such occasions. Small pieces, used to close the ragged end of a tube (see p. 35), allow the two halves of a joint, that fit perfectly, to be brought together with the rubber still between them. Then as each joint is made, the thin sheets of rubber can be removed one by one and the minute gaps can easily be closed with the welding rod.

Removal of Strain and Clamping of Apparatus

Strain in the glass, created during the construction of some piece of apparatus, is usually removed by annealing; however, further strain is sometimes set up when a piece of apparatus is clamped to a rigid framework at more than one point. Annealing alone may not remove strain produced in this way—the glass must be softened so that it will flow. With some pieces of equipment it is undesirable to heat anywhere after the whole has been annealed—as, for example, with a

THE ASSEMBLY OF COMPLEX APPARATUS

diffusion pump—and if clamping at more than one point is unavoidable then this clamping must be performed with extreme care. The claws of the clamp should be well padded with cork, asbestos rope or layers of asbestos paper. If rubber is used it should be dusted with french chalk to prevent sticking which can cause fracture when unclamping at some later date. See p. 39 for remarks concerning types of clamps.

Wherever possible, when a glass tube has been clamped rigidly at two points, the tube between these points should be heated to the softening temperature—so that the glass will flow and take up any inaccuracy in the clamping—and then annealed. In this way an almost completely strain-free system can be obtained. Clearly a rigid framework is necessary if this strain-free condition is to be maintained.

Annealing by Flame

When an annealing oven is not available, even quite large pieces of equipment can be annealed by flame. Apparatus assembled on the bench can often only be annealed by a flame. As a general principle each joint should be annealed as it is completed, unless a further join is to be made, or another operation is to be performed immediately in the same region or so near to it that the whole of the first joint can be kept hot—then the whole is annealed together.

As large a flame as possible is used, with the oxygen adjusted so that the inner blue cone is long and a very pale transparent blue (for *Pyrex* and similar glass). The glass is warmed slowly—with large equipment, made with tubing greater than 40 mm diameter, warming should be begun with a luminous flame or even with hot air—until the whole has been raised to a dull red heat and the glass is distilling sodium (colouring the flame yellow) freely. As much of the glass as possible is kept at this temperature for a few minutes, and then, by slowly reducing the oxygen content and, finally, the size of the flame, the glass is allowed gradually to cool. The flaming is continued until an even, black deposit of carbon is formed.

Use of a Bent Blowpipe and Double Tipping Device

For getting at tubes placed awkwardly in complicated apparatus a bent tubular blowpipe is very convenient. A useful design is shown in FIGURE 83, *I*.

We think that a long thin blowpipe, as in FIGURE 83, *I*, but bent at the end through a greater angle—about 150°—would be very useful for working on complex apparatus. It would enable the back of tubing to be worked from the front.

170

USE OF BENT BLOWPIPE AND DOUBLE TIPPING DEVICE

A double tipping device, which forms two fine flames directed towards one another from nozzles about 2 cm apart, is shown in FIGURE 83, *II* (see p. 29). This blowpipe head is very handy for working small tubing in awkward positions. When joins have to be made in such tubes it is desirable to arrange things so that the ends can be pushed together whenever possible. The use of welding rod at long range, or when the operator has to assume a tedious position, requires a very steady hand.

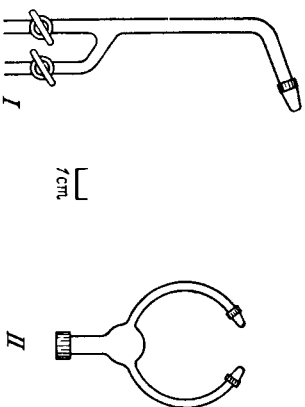


FIGURE 83. Specially shaped blowpipe heads

Danger of Flame Cracks

Recognition of Flame Cracks

Care must be taken, when working on complicated apparatus, to avoid producing flame cracks. These are caused by momentary playing of a hot flame on cool glass. Cracks formed in this way may not appear for several hours, and in some cases even days, after being caused. They generally appear in the surface or bulk of the glass in one of the two forms shown in FIGURE 84. The first is a small curling crack, usually between 5 and 10 mm diameter if circular, and is very characteristically a flame crack. The second general type has a starred appearance with small cracks radiating from a tiny curl or point. The exact mode of formation of these patterns is uncertain, but it is clear that some form of differential strain is the cause. Sudden heating of the surface layer makes it strain-free, and, on cooling, tensile stress develops in the surface layer because the inner layers of the glass are rigid. Glass in tension can undergo a delayed fracture (p. 10).

Precautions for avoiding Flame Cracks

With intricate apparatus, particularly when a hand torch is being

171

used, it is sometimes impossible to avoid playing the flame on to some portion of the assembly that is not actually being worked. Such portions must be protected by wrapping with asbestos paper; this is particularly important when a joint such as an internal seal is in danger. When a hot flame has been inadvertently played on a piece of cool glass, the latter should be annealed, after the work has been completed, to avoid subsequent formation of a flame crack.

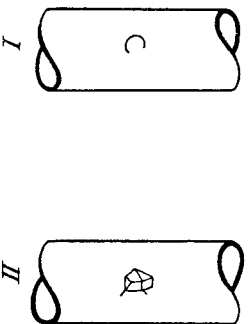


Figure 84. Flame cracks

Removal of Flame Cracks

Flame cracks are easily removed by warming the glass, gently at first to avoid spreading the crack, and raising it to the annealing temperature. When joints are in the vicinity care must be taken to heat everything uniformly and to anneal as a whole.

Chapter 10

THE MANIPULATION OF SILICA

The Properties of Fused Silica and General Remarks

VITREOUS silica, commonly known as *Virreosil*, is pure homogeneous fused silica containing over 99.8 per cent SiO_2 . This glass has many extremely useful properties—such as chemical resistance, extremely low thermal expansion with resultant constancy of volume and resistance to thermal shock, resistance to heat and transparency to ultra-violet and infra-red radiations—making it a valuable material for many scientific purposes. These properties are dealt with in varying detail in Chapter 2. Resistance to heat is often of considerable importance since *Virreosil* has an extremely high softening range, in the region of 1500°C , and can, under normal working conditions, be safely kept at temperatures of 1100°C and as high as 1300°C for short periods. In vacuum work, for evacuated furnaces for instance, fused silica can be used at temperatures up to 1100°C ; and provided no devitrification has occurred the glass remains practically impermeable to gases at these high temperatures, with the possible exception of helium.

Fused silica is manufactured in both translucent and clear, transparent forms; the particular requirements govern the form to be used. The translucent variety can be obtained with a sandy, satin or glazed finish. An interesting historical account of the early work on the practical manipulation of fused silica has been given by W. A. SHENSTONE (1902).

Devitrification is a factor requiring particular attention. Impurities on the surface of *Virreosil* will readily cause crystallization of the surface glass when it is worked. Salts of sodium, potassium and lithium, salts of vanadic acids, fluorides, phosphates and radioactive substances, are especially active in this respect. It is therefore very important, particularly when it is required that the silica remains clear and transparent, that the glass be washed carefully, inside and out, before working. If it has been necessary, during preparation, to grind the fused silica, then all traces of abrasive must be removed by dipping in dilute hydrofluoric acid followed by careful rinsing. This treatment, with subsequent heating to just below the softening point

THE MANIPULATION OF SILICA

with a flame, is also effective in removing any condensed surface deposits of silica due to volatilization during working. Precautions advisable when using hydrofluoric acid are mentioned on p. 158.

Any cleaned portion of the *Vitreosil* must not be touched with the fingers before being worked. When using a bench flame, if the part being handled is subsequently to be worked, the piece of equipment must be re-cleaned before this further work is carried out. For this, cotton-wool dipped in alcohol is very useful—after the piece has been allowed to cool.

General Technique for Working Fused Silica

Cutting Tubing

Fused silica is slightly more brittle than borosilicate glasses and is therefore more easily fractured. Tubes up to just over 1 cm in diameter can be snapped with the fingers after scratching with a file or glass knife. Any ragged end so formed can be cleaned up with an abrasive wheel.

With larger tubing a cutting wheel is necessary since the low thermal expansion renders local heating entirely ineffective in starting a crack. A silicon carbide cutting wheel, $\frac{1}{8}$ to $\frac{3}{8}$ inch thick and running with a peripheral speed of 6000 feet per minute, is very suitable and quite adequate. A copious supply of water directed on the cutting edge is necessary and should be directed from either side of the cutting wheel. Copper or phosphor-bronze, bonded, diamond, cutting wheels can also be used but are much more expensive items.

Fusing and Working Operations

As a medium for glass-working, fused silica has the considerable advantage over the softer glasses that careful and uniform heating and annealing are unnecessary. Slightly offsetting this advantage is the necessity for careful cleaning, as described above, and for stronger heating.

An ordinary oxy-coal-gas torch is only adequate for small size equipment; when working with tubing of greater diameter than 15 mm, oxy-hydrogen or other very hot flames must be used. Special burners are obtainable however—as, for example, the No. 8 *Thermal Syndicate* torch—with which oxy-coal-gas, or, better, oxy-propane mixtures, will give a sufficiently hot flame for this larger tubing, so avoiding oxy-hydrogen mixtures which are less safe in operation unless anti-burn-back valves are employed.

In all fusing operations with fused silica, protective spectacles must be worn—a glass which absorbs at least 96 per cent of the incident

GENERAL TECHNIQUE FOR WORKING FUSED SILICA

light is necessary. The recommended glass for best protection is green *Protex* grade C, $\frac{3}{8}$ inch thick (Chance Bros. Ltd).

For hand work with a bench torch, tubing is rotated in the flame to secure even heating and hence even shaping; while when clamping of the various tubes is employed and the hand torch is used, all the dangers of cracking, which are present when borosilicate or soft glasses are being worked, are negligible. The *Vitreosil* must be blown while still in the flame; and therefore, when a bench torch is used and the apparatus is manipulated by hand, a blowing tube must be attached at one end. It is preferable then to use a glass-blower's swivel (see p. 42).

For tooling of molten *Vitreosil* $\frac{3}{8}$ inch *Nichrome* wire may be employed; though if much tooling is required, then carbon rods or plates are necessary. In general, most of the operations described in earlier chapters can be modified for working with *Vitreosil* and, if a sufficiently hot flame is available, less manipulative skill is usually necessary because the material stays fairly viscous.

When operating with tubes it is well to take care to prevent the flame licking the insides; the vapours can cause contamination which results in devitrification when more work is done.

Large Diameter and Thick-Walled Tubing

With very large silica tubing, over 4 cm bore, the walls are usually very thick. Special methods are required when the walls are 5 mm thick or more. The wall edges, at the ends to be joined, are cut or ground to sharp knife-edges, so that a V-shaped notch is left all round when the ends are butted together. The ends are fused together, using a hand torch and 3–5 mm welding rod to fill in the V-shaped notch. With very large tubes the flame must be manipulated to keep the tubes hot all round, maintaining an even heat, and the apparatus must be annealed—preferably in a furnace for one hour at 1050°C.

Translucent and transparent *Vitreosil* rods of the proper size for the above operations are obtainable.

Silica Torston Fibres

These fibres can be made by drawing by hand or by simple mechanical means, and by blowing them out with a flame. Here we describe only simple methods for making fibres by drawing them out with a weight falling under gravity. With these methods fibres with a range of sizes can be made quite quickly. T. C. KEELEY (1924) fixed a small weight to a silica rod and heated this at a point until the silica softened sufficiently for the weight to fall and draw out a fibre. V. J. CLANCEY (1950), during an investigation of the evaporation and

THE MANIPULATION OF SILICA

surface mobility of naphthalene molecules, used a delicate torsion apparatus, the fibre for which was made by the gravity method. He fused a silica hook to a 1-mm diameter silica rod. The rod was clamped vertically and a weight of about 10 gm hung on the hook. The rod was heated with a flame just above the hook, and just as the weight started to fall the flame was whisked away. The weight fell on to a soft material, such as a rubber sheet, to stop it bouncing. By varying the weight and the moment the flame is taken away, fibres with different sizes can be made. Some preliminary practice is re-

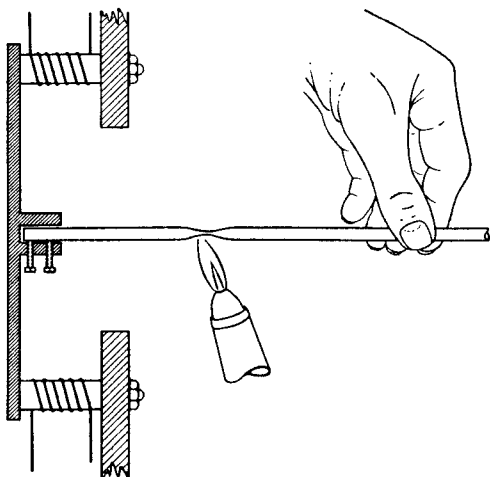


Figure 85. The production of a silica fibre by drawing out a fused rod with a weight falling under gravity

quired. If the flame is kept on too long the fibre will fuse and break, and if the flame is kept on for too short a time the weight will not fall the full length.

We find it more convenient to draw out fibres by the gravity method with a weight of about 500 gm which is released from a pair of electromagnets. The weight consists of an iron bar about 25 cm long, to the centre of which the silica rod of 2–5 mm diameter is attached with a socket and bolt arrangement. The bar is held at each end by a small magnet, and the two magnets are connected in series. A portion of the rod is heated with a hand torch and the top of the rod is held in one hand (Figure 85). A semi-molten portion of

SILICA TORSION FIBRES

desired thickness is obtained, the magnet current is turned off with a foot-operated switch, the flame is whisked away, and the weight falls into a box of sand. If all goes well a long fibre is produced. The semi-molten part of the silica rod, when initially thin, gives a thin fibre, and when thick, a thick fibre; the fibre thickness is also determined by the precise moment the flame is whisked away—for a thin fibre this should be just after the weight starts to fall. Fibres down to 5–10 microns diameter can be made in this way.

N. J. TIGHE (1956) has given a comprehensive survey of the production, properties and applications of silica fibres; G. A. DOWNSBROUGH (1937) has discussed the damping of fibres. With a rubber catapult method, introduced by R. THRELFALL (1898), fibres in the range of 2–30 microns diameter can be made. Smaller fibres can be made by the drag of the flame gases in an oxy-hydrogen flame on a tiny globule of molten silica, a method which Threlfall describes. This author also gives an account of the bow-and-arrow method of C. V. Boys.

Small fibres should be manipulated against a dark—preferably black—background, and small camel's-hair brushes moistened with water are very convenient implements. Fibres, when examined under a microscope, become more easily visible if they are mounted under a cover slip and a drop of glycerine, coloured by a dye, is allowed to run in.

P. L. KIRK and R. CRAIG (1948) have discussed the construction of fibre devices; L. WALDEN (1937) has given a practical account of several uses of fibres as instrument suspensions; D. R. BARBER (1930) has described a device for mounting fibres with a degree of tension; and H. V. NEHER (1940) has given a very practical account of the manipulation of silica fibres.

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AUTHOR INDEX

- Anhorn, V. J., 5, 28, 41, 47, 123, 136, 156, 164
 Ansley, A. J.,⁵
 Armstrong, H. E., 3, 5
 Aston, F. W.,⁴
- Baker, E. A., 87, 89
 Barber, D. R., 177
 Barkas, W., 152, 164
 Barr, W. E., 5, 28, 41, 47, 123, 136, 156, 164
 Berzelius, J. J.,^{3, 5}
 Bodenstein, M.,⁴
 Bolas, B. D.,^{3, 5}, 48, 89
 Bolas, T.,^{3, 5}
 Boyle, R.,⁷, 17, 26
 Boys, C. V.,³, 83, 84, 89, 177
 Braddick, H. J.,⁵
 Brickwedde, F. G., 85, 89
 Brown, B., 84, 89
 Bunsen, R. W.,³, 17, 48, 163
- Callendar, H. L.,^{3, 5}
 Case, F. A., 85, 89
 Chattaway, F. D., 87, 89
 Clancey, V. J., 175, 177
 Cockett, A. H., 161, 164
 Coffin, C. C., 110, 116
 Cook, J. W., 85, 89
 Craig, R., 177
- Dale, A. E., 9, 20, 26
 Daniels, F., 156, 164
 Davies, J., 135
 Dewar, J.,⁴, 123, 164
 Donal, J. S., 85, 89
 Douglas, R. W.,⁹, 26
 Downstrough, G. A., 177
 Dunoyer, L., 139, 164
- Edwards, J. D., 3, 5, 19, 26
 Elliott, A.,⁵
- Faraday, M., 1, 2, 3, 5, 14, 18, 26, 48, 83, 84, 89
 Farkas, A., 149, 164
 Flinders Petrie, W. M., 1, 5
 Foord, S. G., 159, 164
 Fray, F. C.,^{3, 5}, 19, 26
 French, E. A. H., 87, 89
- Gardner, I. C., 85, 89
 Garner, W. E.,^{7, 26}
 Grodzinski, P., 84, 89
 Gunney, C., 11, 26
- Harcourt, W. V.,¹
 Heatley, N. G., 84, 89
 Heidt, L. J., 156, 164
 Heldman, J. D.,^{3, 4, 5}
 Herington, E. F. G., 87, 89
 Home Dickson, J.,⁵
 Housekeeper, W. G., 109, 113, 114, 116
- Jackson, H., 131
 Johansen, A., 15, 26
- Keeley, T. C., 175, 177
 Kirk, P. L., 177
 Kon, G. A. R., 132, 164
- Langmuir, I., 4, 123, 164
 Lavoisier, A. L.,¹
 Lloyd, J. T., 77, 78, 89
- MacDonald, D. K. C., 17, 26
 McLeod, H., 136, 164
 Melville, H. W., 149, 164
 Morey, G. W.,^{1, 6}
 Morgan, E. H.,^{5, 6}
 Morley, A., 119, 165
- Neher, H. V., 177
 Nokes, M. C.,^{5, 6}
- Ockenden, F. E. J., 87, 89
- Park-Winder, W. E.,^{5, 6}
 Partridge, J. H., 15, 27, 44, 47, 102, 111, 116
 Pearson, A. R., 160, 165
 Pearson, S., 11, 26
 Phillips, C. J.,^{9, 27}
 Phillips, N. W. F., 156, 165
 Pollard, J., 145, 146
- Rae, W. N., 151, 165
 Ramsay, W.,⁴
 Randall, J. T., 8, 27
 Reilly, J., 151, 165

AUTHOR INDEX

- Reimann, A. L., 4, 6, 43, 47
 Rideal, E. K., 4, 6
 Roscoe, H. E., 4, 6
 Rosenblum, S., 77, 89
 Scheele, K. W., 1
 Scott, R. B., 85, 89
 Shensstone, W. A., 3, 4, 6, 30, 47, 173, 177
 Smith, I. C. P., 29, 32, 47, 51, 89
 Stanworth, J. E., 9, 17, 20, 26, 27
 Steacie, E. W. R., 156, 165
 Strong, J., 5, 6
 Tammann, G., 7
 Taylor, C. S., 3, 5, 19, 26
 Taylor, G. F., 110, 116
 Thomas, J. S. G., 160, 165
 Threlfall, R., 3, 4, 6, 28, 46, 47, 177
 Threlfall, R. E., 2, 6
 Tiehe, N. J., 13, 27, 177
 Todd, B. J., 17, 27
 Travers, M. W., 4, 6
 Turner, W. E. S., 17, 27
 Upton, P. B. G., 87, 89
 Walden, L., 177
 Walen, R., 77, 89
 Waran, H. P., 144, 145, 165
 Warren, B. E., 8, 9, 27
 West, W., 155, 156, 165
 Wilkinson, R. A., 49, 89
 Wright, R. H., 3, 6
 Yarwood, J., 87, 89
 Yorke, S. G., 159, 165
 Zachariassen, W. H., 8, 27

SUBJECT INDEX

- Abrasives, 39
 Alkaline attack on glass, 16
 Alumina as abrasive, 39
 Ampoule sealing, 29
 Ampoules, 117
 Annealing of glass, 20, 170
 Annealing oven, 45
 Annealing temperature, 12
 Asbestos gloves, 37, 90, 92
 Asbestos, uses of, 36
 Bellows for blowing, 30
 Bellows, glass, 101
 Bench flame method of work, 2
 Bends, 53
 Bends in big tubing, 90
 Blowers for air, 30
 Blowing tubes, 35, 56
 Blowpipes, 28, 170, 174
 Boats, glass, 128
 Borate glasses, 9
 Boring, 72
 Boron atoms, coordination of, 8
 Borosilicate glass composition, 9
 Bottles, cutting big ones, 51
 Bourdon gauge, 159
 Breaking glass tubes, 49
 Breaking of glass under tension, 9
 Breaking of glass when heated, 12
 Break-tip seals, 119
BTB No. 9 glass, 22, 105, 112
BTB No. 12 glass, 24
BTB No. 14 glass, 105
BTB No. 40 glass, 23, 112
BTB No. 93 glass, 85
BTB No. 94 glass, 21
 Bubbles, glass-blowing of, 59
 Bulbs at the end of a tube, 71
 Bulbs, holes in, 72
 Bulbs in a tube, 70, 100
 Burns, treatment when slight, 46
C, 9 glass, 22, 105, 112
C, 40 glass, 23, 112
 Capillary tube working, 68
 Carbon tools, 33, 72, 175
 Carborundum, 39, 44, 82, 83
 Carborundum grinding wheel, 45
 Carus tubes, 118
 Cements for glass, 84
Certrouge, 40, 83
 Chemical properties of glass, 15
 Chromic acid for cleaning, 49
 Claisen flask, 120
 Clamped apparatus, delayed fracture of, 11
 Clamped apparatus, working on, 4, 169
 Clamps, 39, 170
 Cleaning glass tubes, 48
 Closed systems, 167
 Closing a large flask, 92
 Closing tubes with roundends, 57
 Closing tubes with rubber caps, 34, 168
 Closing tubes with stoppers, 35
 Cold finger traps, 125
 Complex apparatus, 166
 Condenser, double surface, 133
 Condenser, fractional distillation, 131
 Condenser, Liebig, 76, 129
 Condensers, 128
 Copper, deposition on glass, 87
 Copper-to-glass seals, 113
 Copper-tungsten-nickel wires, 107
 Corks, 35
Corning No. 3320 glass, 105
Corning Nos. 7040, 7052 and 7060 glasses, 112
Corning No. 7720 glass, 23, 105
Corning No. 7740 glass, 105
 Corundum, 39
 Cracking-off burner, 29
 Cracks, flame, 171
 Cracks, mending, 80, 96
 Crossfire burners, 29
 Crucibles, vitreous silica, 17
 Cut-offs, mercury, 135
 Cutting a tube internally, 38
 Cutting a tube with a flame, 56
 Cutting a tube with a wheel, 44
 Cylinder heads, 30
 Delayed fracture of glass, 10, 11
 Density of glass, 14
 Devitrification of glass, 18
 Devitrification of silica, 173
 Dewar seal, 101, 107, 121
 Dewar vessels, 86, 123
Dial 36 glass, 22, 105

SUBJECT INDEX

- Dial 43* glass, 24, 112
- Dial 444* glass, 23, 108
- Diamond powder as abrasive, 39, 83
- Diamonds, use of, 38, 84
- Diffusion through glass, 17
- Disc seals, copper, 115
- Discharge tube lamps, 154
- Distillation flasks, 120
- Distillation splash head, 121
- Distinguishing types of glass, 26
- Double tipping device, 171
- Drilling holes in glass, 83
- Dumet* alloy wire, 111
- Elastic properties of glass, 13
- Electrical resistance of glass, 14
- Electrodes in apparatus, 156
- Electrolytic gas generator, 149
- Emery, 39
- Equipment required for different purposes, 46
- Extension of glass under tension, 13
- Extraction apparatus, Soxhlet, 142
- FCN* glass, 23, 112
- Feet for vessels, 78
- Ferrio*, 111
- Fibres, delayed elastic effects in, 13
- Fibres, making silica ones, 175
- File, triangular, 38
- Filter pumps, 126
- Fire-polishing, 13
- Firmasil* glass, 22
- Flame annealing, 170
- Flame cracks, 171
- Flame gases, reaction with glass, 17
- Flamemaster* hand torch, 29
- Flanging, 72
- Flaring, 72, 76, 100
- Flaring tools, 31
- Flask clamps, 32
- Flow meters for gas, 152
- Forceps, 32
- Fractional distillation condenser, 131
- Fractional distillation receiver, 132
- Fractionating column, 132
- Frozen surfaces, 82
- Gas analysis apparatus, 163
- Gas generator, electrolytic, 149
- Gas flow meters, 152
- Gauge, Bourdon, 159
- Gauge, glass spiral, 159
- Gauge, McLeod, 76, 136
- Gauges for measuring glass, 34
- Glass, defined, 7
- Glass holders, 40
- Gloves, asbestos, 37
- Gold, firing on to glass, 87
- Graded seal, 101
- Graded seal glasses, 25
- Grinding glass, 40, 45, 82, 141
- Grinding wheel, 45
- Hand torch, 29, 171
- Hand torch, value of, 4
- Handle for tubing, 55
- Hardness of glass, 14
- Heating of glass, effects of, 16
- Helium, diffusion through glass, 17
- Holders for tubes, 40
- Holes, blowing of, 59, 72
- Holes, drilling of, 83
- Holes, filling of with rod, 64
- Holes, finding of, 81
- Hot spot for breaking tubes, 50
- Housekeeper seal, 109, 113
- Hydrofluoric acid, for removing surface layer, 19
- Hydrofluoric acid, precautions with, 158
- Hydrogen discharge lamps, 154
- Hydrophobic glass surfaces, 18
- 'In-place' glass-blowing, 3
- Internal seals, 73, 100
- Internal seals, large, 95
- Jeweller's rouge, 40
- Joints between two big tubes, 93, 100
- Joints between two similar tubes, 60, 100
- Joints between tubes of different size, 64, 100
- Joints, borosilicate glass to vitreous silica, 25
- Joints, borosilicate to soda glass, 25
- Joints in closed systems, 168
- Joints, *Pyrex* to *Pyrex*, 25
- Joints, tungsten-sealing to *Kovar*-sealing glasses, 24
- Joints, *Vitreosil*, 175
- Joints with capillary tube, 69
- Junctions, 4 and 5 way, 68
- King's College condenser, 131
- Knives, glass-cutting, 37
- Knocking off bits of tubing, 51
- Kodial* glass, 23, 112
- Kovar* alloys, 111
- Kovar*-to-*Pyrex* seals, 102, 112
- L.I.* glass, 23
- Lamps, discharge, 154
- Lampworker, 28
- Lapping wheel, 44
- Large tube supports, 41
- Lathe, glass-working, use of, 97, 113
- Lead glass, 23, 108
- Lead glass composition, 9
- Leaks, finding of, 81
- Leaks, molecular flow, 158
- Lemington FCN* glass, 23, 112
- Lemington H.26X* glass, 24
- Lemington W.1* glass, 22, 105, 112
- Light transmission through glass, 15
- Liquid air traps, 125
- Lower annealing temperature, 11
- Lubrication of tools, 32
- M.6 glass, 24
- Machine for glass-working, 97
- McLeod gauge, 76, 136
- Mercury lamps, 155
- Mercury, traps for, 140
- Mercury vapour pumps, 144
- Mesh size for particles, 40
- Metal layers on glass, 87
- Metal-to-glass seals, 102, 156
- Meters for gas flow, 152
- Mg point, 11
- Mica windows fused to glass, 85
- Monax* glass, 21, 118
- Monax* glass, ease of working, 26
- Mouthpieces for blowing tubes, 36
- Network-forming ions, 8, 9
- Network-modifying ions, 8, 9
- Nicosel*, 111
- Nilo*, 111
- Nonex* glass, 23, 105, 112
- Optical properties of glass, 15
- Palladium, firing on to glass, 87
- Perturbations in bulbs, 72
- Phoenix* glass, 22, 105, 106
- Phosphorus pentoxide traps, 127
- Physical properties of glass, 9
- Pinch seals, 106
- Pinholes, 64, 74, 77
- Platinizing glass, 87, 110
- Platinum, sealing into glass, 108
- Polishing glass, 83
- Potassium, contamination of in glass, 17
- Prince Rupert's drops, 20
- Pulling a point, 57
- Pulling a spear, 57
- Pump, filter, 126
- Pumps, circulating, 160
- Pumps, mercury vapour, 144
- Pyrex* glass, 22, 105, 118
- Reaction vessels of glass, maturing of, 18
- Reamers, 31
- Resistance, chemical, of glass, 15
- Resistance, electrical, of glass, 14
- Ribbon burners, 29
- Ring seals, 73
- Rod, thin glass, 58, 64
- Rollers, 42
- Rotation of tubes in the flame, 52
- Rubber caps, 34, 168
- Rubber stoppers, 35
- Sand for grinding, 39
- Scissors for cutting hot glass, 52, 92
- Sealed tubes for reactions, 118
- Sealing glasses, 22
- Sealing glasses for tungsten, 105
- Seals, multiple wire, 105
- Semi-micro analysis apparatus, 162
- Silica fibres, 13, 175
- Silica, properties of, 25, 173
- Silica, working of, 174
- Silvering glass, 85, 87
- Sintered discs, sealing in, 79
- Soda-boric oxide glass, 8
- Soda-lime-silica glass composition, 9
- Soda-silica glass, 8
- Soda-to-*Pyrex* seal with copper tube, 115
- Sodium Resistant NA.10* glass, 24
- Softening temperatures, 11, 13
- Soldering glass after silvering, 88
- Soxhlet extraction apparatus, 142
- Spectacles for glass-working, 37, 174
- Spectrum tubes, 153
- Spike, 31
- Spindles, 56, 91, 99, 100
- Spinning, 73, 78
- Spiral gauge, 159
- Spirals, 53
- Splash head for distillation, 121
- Strain in clamped apparatus, 169
- Strain in glass, thermal, 19
- Strain-viewer, 15, 43
- Strength of glass, 9, 10
- Stress-optical coefficient, 15
- Structure of glass, 7
- Supports for large tubes, 41, 42
- Surface properties of glass, 18
- Swivel connection, 42, 70, 99
- T-joints, 65
- T-joints, capillary, 69
- T-joints, large, 94
- Table for glass-blowing, 28
- Tapers in tubes, 56, 65, 79, 101
- Telcoval*, 111
- Tensile strength of glass, 10
- Tetrahedra of SiO_4 , linking of, 7
- Thermal capacity of glass, 14
- Thermal conductivity of glass, 13

SUBJECT INDEX

- 182
- 183

SUBJECT INDEX

- Thermal endurance of glass, 12
 - Thermal expansion of glass, 11
 - Thermal strain in glass, 19
 - Thermostat regulators, 150
 - Tools for glass-working, 30, 33, 73
 - Tools with glass-working machine, 98
 - Torsion fibres, silica, 175
 - Transformation point, 11
 - Traps, liquid air, 125
 - Traps, phosphorus pentoxide, 127
 - Traps to catch mercury, 140
 - Tubes, holding, 41
 - Tungsten, nickel tipping of, 108
 - Tungsten, sealing to *Pyrex*, 103
 - Upper annealing temperature, 11
 - Uranium glass, 105
 - Vacuum flasks, 123
 - Valves, 140
 - Viscosity of glass, 11, 13
 - Vitreasil*, 25, 173
- W.1* glass, 22, 105, 112
 - Water, evolution from glass, 16
 - Wax for tools, 32
 - Waxes for joining glass, 84
 - Weathering of glass, 16, 18
 - Welding rod, 58, 64
 - Wembley 'Amber Neutral'* glass, 24
 - Wembley L.1. Lead glass*, 23
 - Wembley M.6. 'White Neutral'* glass, 24
 - Wembley X.8.* glass, 21
 - Wheel for cutting glass, 44, 51
 - Wheel for grinding, 45
 - Wheel for lapping, 44
 - Windows, mica, 85
 - Windows, thin glass, 77
 - Wire holders, 104
 - Working various types of glass, 105
 - X.8.* glass, 21
 - X-rays, scattered by glass, 8
 - Y-joints, 68